

Finite Type Invariant of knots

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Abstract

This thesis aims to give a self-contained introduction to the fundamental theorem of finite type invariants of knots. The theory of finite type invariants was developed in 1990s with the work of V. Vassiliev, M. Goussarov, D. Bar-Natan, M. Kontsevich, J. Birman, X.S. Lin. The fundamental theorem of finite type invariants relates them with the algebra of chord diagrams and identifies finite type invariants with weight systems, and was first proved by M. Kontsevich[19] using an integral comes from Knizhnik-Zamolodchikov equation, which is universal among all finite type invariants. Later on, based on the work of many people including Bar-Natan[4] Guadagnini, Martellini, Mintchev[16], Kontsevich[19], Axelrod and Singer[3], Bott and Taubes[8], Dylan Thurston[28] and Altschuler and Freidel[3] independently constructed another universal finite type invariant, the Chern-Simons series which is a power series of configuration space integrals. We will give an elementary introduction to the construction of these invariants and use them to prove the fundamental theorem.

Key words: Finite type invariants, Kontsevich integral, Configuration space integral.

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Chapter 1

Introduction

Historically, finite type invariants were first introduced in [29] by V. Vassiliev who was studying the cohomology of the space of immersions at that time. Vassiliev's original idea was to consider the space of knots $\mathfrak{K} := \text{Emb}(\mathbb{S}^1, \mathbb{S}^3)$ as the complement of “discriminant set” Σ consists of curves that are not embeddings in the space of closed curves $C(\mathbb{S}^1, \mathbb{R}^3)$, and $\pi_0(\mathfrak{K})$ is the set of knots modulo isotopy. Using Alexander duality, the cohomology of $\mathfrak{K}$ is reduced to the computation of $H(\Sigma)$. Then the discriminant set Σ has a filtration coming from the singularities of singular knots, which allows us to define a spectral sequence that converges to the cohomology of $\text{Emb}(\mathbb{S}^1, \mathbb{S}^3)$. The E_1 term then consists of finite type invariants. We will not discuss Vassiliev's original constructions further in this thesis.

The theory of finite type invariants was later decoupled from Vassiliev's sophisticated theory. D. Bar-Natan was the first to give a comprehensive treatment of this theory in his PhD thesis [6], which related them with the topological quantum field theory. Finite type invariants were successively studied by J. Birman and X.S. Lin [7] and M. Kontsevich [19]. It was shown that these invariants are naturally related to combinatorics of algebra of chord diagrams. Every finite type invariants give rise to a functional on the algebra of chord diagrams, called a weight system. In particular, Kontsevich constructed a mysterious integral Z_{KZ} (The Kontsevich Integral) in [19] that turns out to be a universal finite type invariant that covers all finite type invariants. As a result, the space of the finite type invariants is isomorphic to the associated graded space of weight systems. This result is the fundamental theorem of finite type invariants of knots. The antecedents of the Kontsevich integral are iterated integrals that represent the monodromy of Knizhnik-Zamolodchikov equations in Wess-Zumino-Witten model of conformal field theory. Iterated integrals were introduced by K.T. Chen[13] in the 1970s and were used to construct braid invariants by T. Kohno[18]. The computation of Z_{KZ} is rather complicated, even for the unknot. Based on the work of V. Drinfel, T. Le and J. Murakami were able to give a combinatoric definition of Z_{KZ} using Drinfel associators, which proved that it takes values in $\mathbb{Q}$.

In the meanwhile, the theory of the configuration space integrals coming from Chern-Simons theory was also developing. These integrals were introduced by Guadagnini, Martellini, and Mintchev [16] and Bar-Natan [6] independently, and were studied by Kontsevich [19] later. However, the analytic difficulties of proving convergence and invariance were not worked out in a complete and mathematically rigorous manner. Bott and Taubes [8] paved the way using the compactification techniques developed by Fulton and MacPherson[15] and Axelrod and Singer [3]. Later on, D. Thurston in his undergraduate thesis [28] and Alt-

shuler and Freidel [1] independently generalized Bott and Taubes' results to all orders and constructed another universal finite invariants Z_{CS} which proved the fundamental theorem of finite type invariants once again. It is conjectured that $Z_{CS} = Z_{KZ}$, based on the work of S. Poirier[26], C. Lescop constructed an isomorphism of $\mathcal{A}$ that identifies two universal finite type invariants in [22].

In Chapter 1, we briefly introduce basic definitions of knot theory and classical knot invariants, especially the linking number. Then we set the stage and discuss the theory of finite type invariants and state the fundamental theorem in Chapter 2. In Chapter 3 we construct the Kontsevich integral, and prove it's a universal finite type invariant. In Chapter 4 we review the compactification of configuration spaces and give a sketchy proof of invariance and universality of the Chern-Simons series.

1.1 Knots, Links and Their Invariants

1.1.1 Basic Definitions

We begin by giving some basic definitions about knots and links.

A k -component *link* L is a smooth embedding (i.e. an immersion that is a topological embedding) of a 1-dimensional compact manifold with k components $M \cong \coprod^k \mathbb{S}^1$ into $\mathbb{R}^3$. A *knot* $K : \mathbb{S}^1 \rightarrow \mathbb{R}^3$ is just a 1-component link. For topologists, two links L_1, L_2 are considered to be the same if one can smoothly distort one to another, namely one can connect them through an *ambient isotopy* $h_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3, t \in [0, 1]$ where h_t is a diffeomorphism of $\mathbb{R}^3$ for each fixed t , $h_0 = \text{id}$ and $h_1 L_1 = L_2$. Given a representative of a link $L : \coprod^k \mathbb{S}^1 \rightarrow \mathbb{S}^3$, we can fix a projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ to get a smooth map $\pi L : \coprod^k \mathbb{S}^1 \rightarrow \mathbb{R}^2$. We shall consider only the cases when πL is an immersion with finitely many self-intersections and each double point transverse (the tangent vectors span the plane). Thus one can draw the information of a regular projection diagrammatically on the plane $\mathbb{R}^2$, adding the information of over and under by drawing a crossing on each double point.

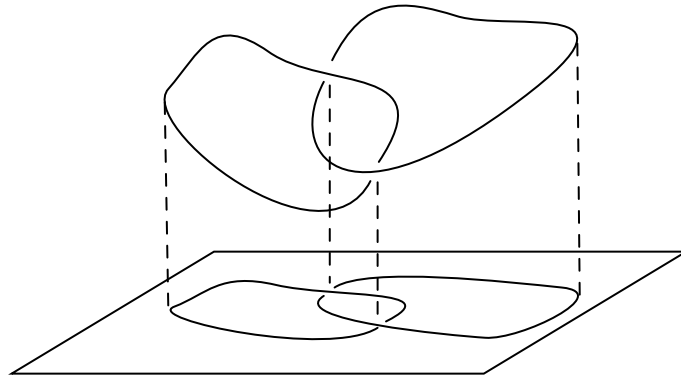
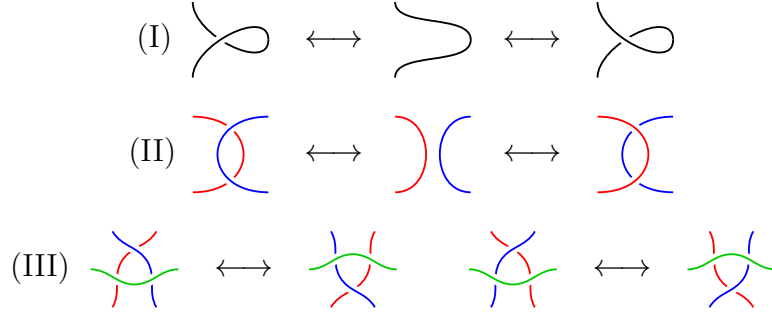


Figure 1.1: A Hopf link and its projection

Let's call such a map a regular projection and the corresponding picture its diagram. It can be proved that every link admits a regular projection. Though a regular projection of a link relies on many choices, in 1926, K. Reidemeister showed that every two regular projections of two representatives of the same link can be related by the following sequence of three types of moves called *Reidemeister moves*:



This allows us to work on knot diagrams directly without loss of any information. The main task of the theory of knots and links is to tell them apart. One way to show two links are the same is to show that a diagram of one of them can be deformed to a diagram of another by a sequence of finitely many Reidemeister moves. However, in general this method will not allow us to distinguish links. Thus, we need to find some easy-to-compute functions on diagrams that are unchanged under Reidemeister moves.

1.1.2 Classical Knot Invariants

We shall now give examples of several most common knot invariants.

Example 1.1.1 (Unknotting Number). Any knot can be unknotted by finitely many crossing changes. We simply define the unknotting number $u(K)$ of a knot K to be the minimum number of crossing changes needed to untie it.

Example 1.1.2 (Jones Polynomial). The *Jones Polynomial* J on a link L can be defined by the Kauffman bracket:

$$J(L) = (-A)^{-3w} \langle L \rangle \quad (1.1)$$

where the writhe w is the signed sum of the number the crossings of the diagram

$$w = \# \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) - \# \left(\begin{array}{c} \nwarrow \\ \searrow \end{array} \right) \quad (1.2)$$

and the bracket is defined inductively by

$$\begin{aligned} \left\langle \begin{array}{c} \nearrow \\ \searrow \end{array} \right\rangle &= A \left\langle \begin{array}{c} \nearrow \\ \nearrow \end{array} \right\rangle \left\langle \begin{array}{c} \searrow \\ \searrow \end{array} \right\rangle + A^{-1} \left\langle \begin{array}{c} \nearrow \\ \nwarrow \end{array} \right\rangle \left\langle \begin{array}{c} \searrow \\ \nwarrow \end{array} \right\rangle \\ \langle L \sqcup \bigcirc \rangle &= (-A^2 - A^{-2}) \langle L \rangle \end{aligned} \quad (1.3)$$

J is uniquely determined by initial condition $J(\bigcirc) = 1$ and the skein relation (normalized by substitution $A = t^{-1/4}$)

$$t^{-1} J \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) - t J \left(\begin{array}{c} \nwarrow \\ \searrow \end{array} \right) = (t^{1/2} - t^{-1/2}) J \left(\begin{array}{c} \nearrow \\ \nearrow \end{array} \right) \left\langle \begin{array}{c} \searrow \\ \searrow \end{array} \right\rangle \quad (1.4)$$

Example 1.1.3 (Conway Polynomial). Conway polynomial is uniquely determined by a choice on unknot $C(\bigcirc) = 1$ and Conway's skein relation:

$$C \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) - C \left(\begin{array}{c} \nwarrow \\ \searrow \end{array} \right) = t C \left(\begin{array}{c} \nearrow \\ \nearrow \end{array} \right) \left\langle \begin{array}{c} \searrow \\ \searrow \end{array} \right\rangle \quad (1.5)$$

1.2 Configuration Space

A configuration space of n distinct points on a manifold M is:

$$\text{Conf}_n(M) := \{(x_1, \dots, x_n) : x_i \neq x_j \text{ whenever } i \neq j\}. \quad (1.6)$$

It is a subset of M^n formed by removing “diagonals”, and is not compact if $n \geq 2$.

Example 1.2.1. $G = \mathbb{R}^n \rtimes \mathbb{R}^+ \subset \text{Aff}(\mathbb{R}^n)$ acts on $\text{Conf}_2(\mathbb{R}^n)$ diagonally by translation and dilation. Note that there is a cross-section of G , $\text{Conf}_2(\mathbb{R}^n) = G \times \mathbb{S}^{n-1}$. In particular, the map

$$\Phi_{1,2} : \text{Conf}_2(\mathbb{R}^n) \rightarrow \mathbb{S}^{n-1}, (x_1, x_2) \mapsto \frac{x_1 - x_2}{|x_1 - x_2|} \quad (1.7)$$

is a deformation retraction.

In general, $\text{Conf}_k(\mathbb{R}^n) = G \times \text{Conf}_k(\mathcal{T}\mathbb{R}^n)$, $\text{Conf}_k(\mathcal{T}\mathbb{R}^n)$ is called the relative configuration space. Consider

$$(\mathbb{R}^n)^k \setminus \{(x_1, \dots, x_n), x_1 = \dots = x_n\} \cong G \times \mathbb{S}^{kn-n-1}. \quad (1.8)$$

There is an embedding into the compact space:

$$\text{Conf}_n(\mathcal{T}\mathbb{R}^n) \hookrightarrow \mathbb{S}^{kn-n-1} \quad (1.9)$$

These relative configuration spaces will be needed in the compactification of configuration spaces later.

Another interesting question is to compute the cohomology of the configuration spaces. Using the spectral sequence of

$$\mathbb{C} \setminus \{1, \dots, n-1\} \hookrightarrow \text{Conf}_n(\mathbb{C}) \rightarrow \text{Conf}_{n-1}(\mathbb{C}) \quad (1.10)$$

Anorl computed[2]:

Lemma 1.2.2 (Anorl). *Under the identification*

$$\omega_{ij} = \frac{dz_i - dz_j}{z_i - z_j} \quad (1.11)$$

$H_{dR}^\bullet(\text{Conf}_n(\mathbb{C}))$ is isomorphic to the exterior algebra generated by $\omega_{ij} = \omega_{ji}, i = 1, \dots, n$ subject to Anorl’s identity

$$\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0 \quad (1.12)$$

The forms $dz_i - dz_j / (z_i - z_j)$ will be used to construct connections on the configuration space $\text{Conf}_n(\mathbb{C})$.

1.3 Linking Number

Intuitively speaking, the linking number is a description of how many times one space curve winds around another, and the origin of it can be traced back to Gauss and Maxwell[27]. There are many ways to compute this number, and in some sense, as we shall see in chapter 3 and chapter 4, the two different constructions of universal Vassiliev invariants can be viewed as generalizations of two different formulas of linking number.

1.3.1 Gauss Integral

The simplest definition of the linking number $lk(L_1, L_2)$ of two link components $L_1, L_2 : \mathbb{S}^1 \rightarrow \mathbb{R}^3$ is the degree of the map

$$\Phi_{L_1, L_2} : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \mathbb{S}^2, (\theta_1, \theta_2) \mapsto \frac{L_2(\theta_2) - L_1(\theta_1)}{|L_2(\theta_2) - L_1(\theta_1)|} \quad (1.13)$$

namely, for an orientation form ω on $\mathbb{S}^2$ with $\int_{\mathbb{S}^2} \omega = 1$,

$$lk_G(L_1, L_2) = \int_{\mathbb{S}^1 \times \mathbb{S}^1} \Phi_{L_1, L_2}^*(\omega) \quad (1.14)$$

Since any isotopy of $L_1 \cup L_2$ gives a homotopy of Φ_{L_1, L_2} , this is a link invariant. If we use the canonical $SO(3)$ invariant form

$$\omega = \frac{1}{4\pi} (zdx \wedge dy + xdy \wedge dz + ydz \wedge dx) \quad (1.15)$$

under the standard coordinate (x, y, z) of $\mathbb{R}^3$ and standard imbedding $\mathbb{S}^2 \hookrightarrow \mathbb{R}^3$, eq. (1.14) is the modern interpretation of Gauss's original derivation of linking number in his investigation of electromagnetic theory.

Since the degree is independent of the choice of a form, then a bump form allows us to calculate the degree locally by a signed sum of the pre-images of a regular value. It's easy to see that, after choosing a link diagram obtained from a projection along some direction $e \in \mathbb{S}^2$, the local degree at e is exactly half of the sum of "external" crossings (one of the strands comes from L_1 and one of them from L_2) with signs determined by the right-hand rule:

$$\begin{aligned} lk(L_1, L_2) &= \# \left(\begin{array}{c} \text{blue} \nearrow \\ \text{green} \nwarrow \end{array} \right) - \# \left(\begin{array}{c} \text{blue} \nwarrow \\ \text{green} \nearrow \end{array} \right) \\ &= \frac{1}{2} \left(\# \left(\begin{array}{c} \text{blue} \nearrow \\ \text{blue} \nwarrow \end{array} \right) + \# \left(\begin{array}{c} \text{green} \nearrow \\ \text{green} \nwarrow \end{array} \right) - \# \left(\begin{array}{c} \text{blue} \nwarrow \\ \text{green} \nwarrow \end{array} \right) - \# \left(\begin{array}{c} \text{green} \nearrow \\ \text{blue} \nearrow \end{array} \right) \right) \end{aligned} \quad (1.16)$$

One can also verify that this combinatorial description of $lk(L_1, L_2)$ is coherent with three Reidemeister moves.

By Alexander duality, the first homology group of a knot exterior $H_1(\mathbb{R}^3 \setminus L_1) = H_1(\mathbb{S}^3 \setminus L_1) = H^1(L_1) = \mathbb{Z}$, and the generator of $H_1(\mathbb{S}^3 \setminus L_1)$ is the meridian $[m_1]$ of the boundary of a tubular neighborhood of the knot L_1 . The homology class $[L_2]$ of L_2 in $H_1(\mathbb{R}^3 \setminus L_1)$ satisfies the equation: $[L_2] = lk(L_1, L_2)[m_1]$.

In general, the linking number of two disjoint null-homologous cycles L_1, L_2 embedded in a $(\dim L_1 + \dim L_2 + 1)$ -manifold M can be defined by the algebraic intersection number of L_1 with a chain C_2 such that $\partial C_2 = L_2$ (in our case, C_2 is the Seifert surface). Seifert algorithm allows us to compute this number from a link diagram as in eq. (1.16), so this definition coincides with the degree of the Gauss map in our special case. But we can see a more direct correspondence in any homology sphere S . Suppose we have two link components L_1 and L_2 in $\mathbb{R}^3$, then the Gauss integral is a composition of the natural map

$$L_1 \times L_2 : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \text{Conf}_2(\mathbb{R}^3) \quad (1.17)$$

with the homotopy equivalence $\Phi_{1,2}$ in eq. (1.7), and the Gauss linking number $lk_G(L_1, L_2)$ is determined by the equation

$$(L_1 \times L_2)_*[\mathbb{S}^1 \times \mathbb{S}^1] = lk_G(L_1, L_2)[S] \quad (1.18)$$

in $H_2(\text{Conf}_2(\mathbb{R}^3))$, where $[S]$ is a generator of $H_2(\text{Conf}_2(\mathbb{R}^3))$. For a detailed proof of the equivalence of lk_G and lk in general homology sphere, see [24].

1.3.2 Winding Number

In the complex plane, there is a very convenient way to measure how many times a closed plane curve $\gamma : [0, 1] \rightarrow \mathbb{S}^1 \rightarrow \mathbb{C} \setminus a$ wraps around a fixed point $a \in \mathbb{C}$. Consider the universal covering map $\exp + a : \mathbb{C} = \widetilde{\mathbb{C} \setminus a} \rightarrow \mathbb{C} \setminus a : z' \rightarrow z$, and let $\tilde{\gamma}$ be the lift of γ , then

$$\text{Wind}(\gamma, a) = \frac{1}{2\pi\sqrt{-1}} (\tilde{\gamma}(1) - \tilde{\gamma}(0)) \quad (1.19)$$

is a homotopy invariant called the *winding number* of γ about a . γ is exactly number $\text{Wind}(\gamma, a)$ powers of generator in $\pi_1(\mathbb{C} \setminus a) = \mathbb{Z}$.

We can explicitly compute this number, note that the pullback of closed 1-form $dz/(z - a)$ by $\exp + a$ is $d\tilde{z}$, thus the Cauchy integral

$$\text{Wind}(\gamma, a) = \frac{1}{2\pi\sqrt{-1}} \int_{\tilde{\gamma}} d\tilde{z} = \frac{1}{2\pi\sqrt{-1}} \int_{\gamma} \frac{dz}{z - a} \quad (1.20)$$

Sometimes we denote the coordinate of the covering space $\widetilde{\mathbb{C} \setminus a} \rightarrow \mathbb{C}$ by $\log$, and we write $d \log(z - a)$ instead of $dz/(z - a)$ and call it logarithmic form.

Note also that eq. (1.20) is the complex-analytical formulation of the mapping degree of

$$\mathbb{S}^1 \xrightarrow{\gamma} \mathbb{R}^2 \setminus a \rightarrow \mathbb{S}^1 \quad (1.21)$$

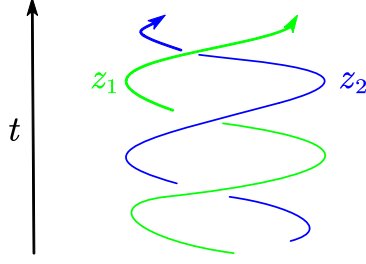
because the form $d \log(z - a)$ is in the same homology class with $d \log(z - a) - d \log |z - a|$. Thus, it's not surprising one can also calculate the winding number locally: we can draw a ray from the origin in the appropriate direction, and count the signed intersection.

If now we have two disjoint (not necessarily closed) curves parametrized by $z_1(t), z_2(t), t \in [0, 1]$, or equivalently a path $I \rightarrow \text{Conf}_2(\mathbb{C}), t \mapsto (z_1(t), z_2(t))$, then we can also use the integral

$$\text{Wind}(z_1(t) - z_2(t), 0) = \frac{1}{2\pi\sqrt{-1}} \int_0^1 \frac{dz_1 - dz_2}{z_1 - z_2} \quad (1.22)$$

to compute the angles z_1 wraps around z_2 . This number is a fixed-end-point homotopy invariant. In particular, if $z_1(1) - z_2(1) = \pm[z_1(0) - z_2(0)]$, we can add another t -axis and “lift” these two curves along t , eventually we get a *2-braid* and the number we just computed is its braiding number, the number of full twists.

Although links can be viewed as analogues of braids (more precisely, tangles) after choosing a certain decomposition $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}_t$, the story for links is more involved and complicated. Braids are traces of points in configuration spaces that can only move forward as t increases, while points on links, can move both forward and backward along the t -axis. This means



that at the same t , there are different choices of pairing. A brutal way to remedy the problem is to simply add everything up, with a careful choice of signs. And indeed, this gives us the following formula of the linking number.

We assume the link components L_1, L_2 are oriented and nicely embedded with respect to the decomposition $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}_t$ in the sense that the restriction of the projection $t : \mathbb{R}^3 \rightarrow \mathbb{R}_t$ on L_1, L_2 is a strict Morse function (has only non-degenerate critical values and all of them are distinct) such an embedding is called a *strict Morse link*(fig. 1.2).

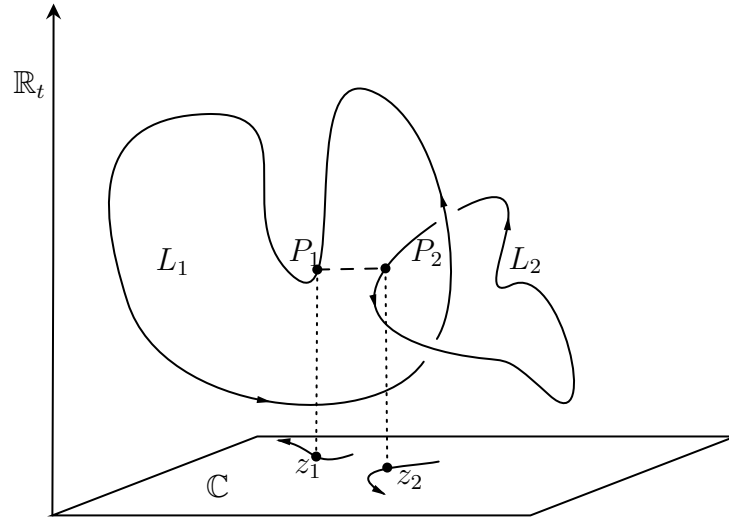


Figure 1.2: Kontsevich Formula of Linking Number

Consider the following configuration space of all possible pairs at the same height (visualized as horizontal chords as in fig. 1.2)

$$C(L_1, L_2) = \left\{ P = (P_1, P_2) \in L_1 \times L_2 (\mathbb{S}^1 \times \mathbb{S}^2) \subset \text{Conf}_2(\mathbb{R}^3), t(P_1) = t(P_2) \right\} \quad (1.23)$$

It's easy to see that this is a smooth 1-manifold by verification on critical points. Being compact, it is a disjoint union of finite many circles. We can define an orientation on $C(L_1, L_2)$ by the orientation form (except on the critical values) $(-1)_P^\downarrow dt$ where $\downarrow_P$ is the number of endpoints in the pair P that is oriented downwards on the link. Let's compute the total winding number of

$$\begin{aligned} \gamma : C(L_1, L_2) &\rightarrow \mathbb{C} \\ P = (P_1, P_2) &\mapsto z(P_1) - z(P_2) \end{aligned} \quad (1.24)$$

$C(L_1, L_2)$ is partitioned by chords at the critical values into n parts $\{\gamma_j\}$, then

$$\text{Wind}(\gamma, 0) = \frac{1}{2\pi\sqrt{-1}} \sum \int_{\gamma_j} d \log(z(P_1) - z(P_2)) \quad (1.25)$$

Substitute P by t , and take orientation into consideration, we immediately get

$$lk(L_1, L_2) = \frac{1}{2\pi\sqrt{-1}} \sum \int (-1)^{\downarrow_P} d \log(z_1 - z_2) \quad (1.26)$$

where the sum is taken over all possible pairs of strings $t \mapsto (z_1, z_2)$. The integral is invariant under horizontal deformations. One can show that the integral is also invariant under moves and insertion or deletion of critical points, just like that we will do to a generalization of this integral in chapter 3. But here we use a simpler proof. As we have mentioned, the winding number of γ can also be computed locally, namely we choose a horizontal direction $N \in \mathbb{S}^1$ and count the signed number of pairs P such that $P_1 P_2$ parallel to that direction. After some suitable horizontal deformation so that the projection of the link along the direction N is regular, again we get the combinatorial description of the linking number: if we project along this direction, this becomes the signed sum of the number of crossings between the two components. Thus the formula eq. (1.26) using logarithmic form coincides with our previous definitions.

1.3.3 What About Knots?

Given a link with components $L_i : \mathbb{S}^1 \rightarrow \mathbb{R}^3$, we have already see that there is a natural map $\Phi_{L_i, L_j} : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \text{Conf}_2(\mathbb{R}^3) \rightarrow \mathbb{S}^2$. Compute the degree of this map and we get the linking number. Given a knot $K : \mathbb{S}^1 \rightarrow \mathbb{R}^3$, one can also form a natural product

$$\Phi_K : \text{Conf}_2(\mathbb{S}^1) \xrightarrow{K \times K} \text{Conf}_2(\mathbb{R}^3) \rightarrow \mathbb{S}^2. \quad (1.27)$$

One may hope the degree

$$I(K, \bigcirc) := \int_{\text{Conf}_2(\mathbb{S}^1)} \Phi_K^*(\omega) \quad (1.28)$$

of this map is a knot invariant, but unfortunately, $\text{Conf}_2(\mathbb{S}^1) \cong \mathbb{S}^1 \times (-1, 1)$ is not compact, so the integral has the problem of convergence. To fix this problem, one can try to compactify the configuration space by attaching two circles to the cylinder, the limit of $\Phi_K(\theta_1, \theta_2)$ is $\pm \dot{K}/|\dot{K}|$ as $\theta_1 - \theta_2 \rightarrow \pm 0$, so we can extend Φ smoothly to $\text{Conf}_2[\mathbb{S}^1] := \mathbb{S}^1 \times [-1, 1]$. But again, this is not an invariant since the space has boundaries, thus the degree might depend on the choice of the form ω , and is clearly not homotopy invariant. For example, if we choose an almost planar embedding (meaning that the knot is embedded into a plane $\mathbb{C} \in \mathbb{R}^3 = \mathbb{R}_t \times \mathbb{C}$ except in the neighborhood of a crossing it lies in some vertical plane and the vertical component of the derivative $\dot{K}$ is small), then $\Phi(\mathbb{S}^1 \times \{\pm 1\})$ is contained in a small neighborhood of the equator. As a result, $I(K, \bigcirc)$ will give the writhe

$$w = \# \begin{array}{c} \diagup \\ \diagdown \end{array} - \# \begin{array}{c} \diagdown \\ \diagup \end{array} \quad (1.29)$$

of the projection onto the plane.

In general, this integral is the weighted average of the writhe obtained from projections along all possible regular directions, and hence will not necessarily be an integer. One can try to use Stokes theorem to see what's wrong and fix the integral by adding a correction term

$$\text{Tw}(K, \mu) = \frac{1}{2\pi} \int \frac{(\dot{K}, \mu, \dot{\mu})}{|\dot{K} \wedge \mu|^2} |\dot{K}(\theta)| d\theta, \quad (1.30)$$

the total twist of (K, μ) . This is a well-known result of J. White, B. Fuller and G. Călugăreanu [11]. We shall not go into details here, but eventually, we get the self-linking number

$$lk(K, K_\mu) = I(K, \bigcirc) + \text{Tw}(K, \mu) \quad (1.31)$$

which is the linking number of the knot K and a copy K_μ of it pushed ϵ away along the framing μ .

Also, one might want to define an integral the same way as eq. (1.26) with logarithmic forms to count the winding number of horizontal chords, except that endpoints sit on the same knot K . But immediately you run into horrible convergence problems as you reach a critical point.

In any case, it seems that things get somewhat tricky when we intend to generalize the linking number to knots. We shall see later that, instead of keeping track of the motion of one single pair, there are some clever ways to define integrals to keep track of finite many pairs on a knot and detect how they rotate.

Chapter 2

Finite Type Invariants of Knots

In this chapter, we study the theory of finite type invariants of knots by some combinatorial methods that relate them to some bialgebras of diagrams. We state the fundamental theorem of finite type invariants which identifies the space of finite type invariants and weight systems, and we will prove the easy part of the theorem. The difficult part (the “Kontsevich” part) will be proved in the next two chapters. Some basic algebraic facts on bialgebras are listed in appendix [A](#).

2.1 Motivation: Do Some Calculus

The main idea of the theory of finite type invariants is, naively speaking, to do “discrete calculus” on the set of knots and “differentiate” knot invariants. The discussions proposed in this section are not necessary to define the finite type invariants of knots, but they are instructive if we want to extend the notion to other categories such as 3-manifolds.

Recall for a function on $\mathbb{Z}$, say $(a_j), j \in \mathbb{Z}$, we can take derivative by take forward difference $\Delta a_j = a_{j+1} - a_j$, or we can take backward difference $\nabla a_j = a_j - a_{j-1}$, and inductively we can define finite differences. We are especially interested in sequences with vanishing differences of order n . These are constants, arithmetic sequences, and their higher order analogues, which have a closed formula as polynomials about j . One can do the same thing for $\mathbb{Z}^d$, except that we have more difference operators $\{\Delta_i, \nabla_i : i = 1, \dots, d\}$. Note also that, these operators can be globally defined because all local neighborhoods are isomorphic, so applying them to a function on $\mathbb{Z}^d$ will again produce a function on $\mathbb{Z}^d$.

From these examples, we learned that to differentiate functions on some set X , we need to decide what is a neighborhood of a point (for $n \in \mathbb{Z}$, this is $n \pm 1$), and usually we do this by deciding a family of operations, $(\pm 1 \text{ for } \mathbb{Z})$. However, if we want to deal with some structures that are not homogeneous, such as arbitrary graphs, we should define differentiation and polynomials more carefully, since there is a vast set of difference operators that only defined locally at every point.

Let’s make this precise. Suppose we have a set $X_0 = X$, and a family X_1 of operations that each $o \in X_1$ sends an element $x \in X_0$ to another element $y \in X_0$, we can visualize this structure as a (possibly infinite) directed graph whose vertices are elements in X_0 and oriented edges are elements in X_1 . We also need a class of distinguished commutative relations of X_1 . Such commutativity is presented by a family of higher dimensional oriented “ n -cubes” whose vertices are elements in X_0 and oriented edges labeled by elements in X_1 .

We shall choose an allowable sub-collection X_n of n -cubes, such that $\bigcup_j X_j$ actually form a *cubic complex*. The general theory of finite type invariants is closely related to cubic structures[25]. There are boundary operators from an n -cube to its i -th faces: $\partial_i^\pm : Q_n \mapsto Q_{n-1,i}^\pm$ satisfying $\partial_i^{\epsilon_1} \partial_j^{\epsilon_2} = \partial_{j-1}^{\epsilon_2} \partial_i^{\epsilon_1}$, $\epsilon_1, \epsilon_2 \in \{\pm\}$ whenever $i < j$. Write

$$\partial_i = \partial_i^+ - \partial_i^- : Q_n \mapsto Q_{n-1,i}^+ - Q_{n-1,i}^- \quad (2.1)$$

Define $\partial^n = \prod_{i=1}^n \partial_i : \mathbb{Z}X_n \rightarrow \mathbb{Z}X_0$. This map does not depend on the order of the product.

In many interesting cases, it's more convenient to use the following equivalent description. For an n -cube $Q_n \in X_n$, there are $n \times 2^{n-1}$ edges, and 2^{n-1} of them in each direction, we shall do not distinguish edges in the same direction and write them $o_1, \dots, o_n$, and regard them as operations that commutative with each other. As a consequence, Q_n is equivalent to the initial vertex x in Q_n along with operations $o_1, \dots, o_n$. We can define the complete resolution map by sending a n -cube in X_n to the signed sum of vertices in X_0 ,

$$\begin{aligned} \partial^n : \mathbb{Z}X_n &\rightarrow \mathbb{Z}X_0 \\ \{x; o_1, o_2, \dots, o_n\} &\mapsto [x; o_1, o_2, \dots, o_n] = \sum_{I \subset \{1, \dots, n\}} (-1)^{\#I} \prod_{i \in I} o_i(x) \end{aligned} \quad (2.2)$$

Now, if we are working on some base field $\mathbb{F}$ of characteristic 0, let $\mathcal{X}_n = \partial^n(\mathbb{F}X_n) \subset \mathcal{X}_0$. Since $\bigcup_n X_n$ is a cubic complex, $\mathcal{X}_n$ is a descending filtration of $\mathcal{X}_0$. An $\mathbb{F}$ -valued function V on X is of *finite type of degree n* , if V vanishes on $\mathcal{X}_{n+1}$. If we view $\text{Hom}(\partial^n, \mathbb{F})$ as a differential operator, and call $V^{(n)} := V\partial^n$ the n -th derivative of V , then V is a finite type function iff its $V^{(n+1)}$ vanishes as a function on X_{n+1} . Let $\mathcal{V}_n$ be the set of all finite type invariants of degree n , then there is also an ascending filtration $\mathcal{V}_n$. $\mathcal{V}$ is the filtered dual of $\mathcal{X}$.

Let's see some examples of the theory of finite type invariants.

Example 2.1.1. $X = \mathbb{Z}^d$, the operations are $\{\pm e_j, j = 1, \dots, d, r \in \mathbb{R}\}$ where $\{e_j\}$ is standard basis of $\mathbb{Z}^d$. The derivative of a function P on $\mathbb{Z}^d$ is defined by

$$\partial_j P(x) = P(x + e_j) - P(x) \quad x \in \mathbb{Z}^d, j = 1, \dots, d, \quad (2.3)$$

it's a polynomial of degree n if and only if all derivatives $\partial_J P(x) = \prod_{i=1}^{n+1} \partial_{j_i} P(x)$, $J = \{(j_1, \dots, j_{n+1}), j_i \in \{1, \dots, d\}\}$ and an n -cube is just a choice of x and multi-index $J = \{(j_1, \dots, j_n), j_i \in \{1, \dots, d\}\}$. Now we give a description of finite type invariants on $\mathbb{Z}^d$, the method used here will be utilized to describe the finite type invariants of knots. To avoid confusion, it is important to note that the notations used in this example are intended for $\mathbb{Z}^d$, but the same notations will be used for knots in the upcoming sections. Let $\mathcal{A} = \bigoplus_n \mathcal{A}_n$ be the algebra of "ascending multi-index" where

$$\mathcal{A}_n = \{J = (j_1, \dots, j_n) \in \{1, \dots, d\}^n, j_1 \leq \dots \leq j_n\} \quad (2.4)$$

with the multiplication defined by juxtaposition. Let $\mathcal{W}_n$ be the space of sequences labeled by ascending multi-index: $\mathcal{W}_n = \{(a_J)_{J \in \mathcal{A}_n}\}$, and $\mathcal{W} = \bigoplus_n \mathcal{W}_n$. The n -th derivative of a degree n function is constant over $\mathbb{Z}^d$ and independent of the order we differentiate it, thus we have a map $\text{Symb}_n : \mathcal{V}_n \rightarrow \mathcal{W}_n, P \mapsto \partial_J P$. But two functions have the same n -th derivative if and only if they differ from each other by a degree $(n-1)$ function, so Symb_n

descends to an injective map $\phi_n : \mathcal{V}_n / \mathcal{V}_{n-1} \rightarrow \mathcal{W}$. This map turns out to be an isomorphism because we can always find a function with prescribed top-degree derivatives. The inverse is given explicitly by a power series in $\hat{\mathcal{A}}$:

$$Z(x) = \exp(x_1(1) + \cdots + x_d(d)) = \sum_J \frac{x_J}{|J|!} J \quad (2.5)$$

where $x_J = x_{j_1} \cdots x_{j_n}$, $|J|! = a_1! \cdots a_k!$ if $J = (j_1, \dots, j_n)$ with a partition $a_1, \dots, a_k$ of n according to the number of different elements. Now $\phi = (\phi_n)_{n \geq 0} : \text{gr} \mathcal{V} \rightarrow \mathcal{W}$ is an isomorphism implies that $\mathcal{W} \rightarrow \mathcal{V}, W \mapsto W(Z)$ is also an isomorphism. But $\mathcal{W}$ is exactly the space of polynomials on d variables. As a consequence, type n invariants are exactly polynomials of degree n on $\mathbb{Z}^d$, and Z is a universal finite type function on $\mathbb{Z}^d$.

Example 2.1.2. $X = \mathbb{R}^d$, the operations are $\{\pm r e_j, j = 1, \dots, d, r \in \mathbb{R}\}$ where $\{e_j\}$ is standard basis of $\mathbb{R}^d$, and the n -cubes are determined by all (possibly degenerate) n -dimensional parallelotopes. Finite type invariants of degree n are polynomials on $\mathbb{R}^d$.

Example 2.1.3. If $X = \mathcal{K}$, the set of isotopy classes of knots, then the operations are crossing changes, and a n -cube is determined by doing crossing changes on n disjoint neighborhoods of crossings of an embedding.

Example 2.1.4. X is the set of integral homology 3-spheres $\mathcal{M}$, and the operations are Dehn surgeries along a framed knot, the n -cubes are determined by Dehn surgeries along an algebraic split and unit framed n -component link.

Example 2.1.5. Let G be a group, an n -cube is a word $g_0(a_1, b_1)g_2 \cdots g_{n-1}(a_n, b_n)g_n$ with $a_i, b_i, g_i \in G$. The boundary operator $\partial_i^+(\partial_i^-)$ takes (a_i, b_i) to be $a_i b_i$ ($b_i a_i$). As a result, two group elements $g - h$ in a n -cube is equivalent to $[g, h]$ is in the n -th lower central group $G_n = [G, G_{n-1}]$.

2.2 The Story For Knots

2.2.1 Finite Type Invariants as Polynomials

For a knot K , there is also a natural operation to perturb it: do crossing changes. Now let's do the same thing about knots as we have done to $\mathbb{Z}$. We construct the cubic complex as in the previous section, with $\mathcal{K}_0 = \mathcal{K}$ be the isotopy classes of knots, and edges connecting any pair of knots that one can be transformed to another by crossing changes. We also need to decide our choice of higher dimensional cubes. Given a knot diagram, we can choose n crossings in n disjoint neighborhoods labeled by a set I , and we use $o_i, i \in I$ to denote the operation of crossing changes on the i -th crossing. Indeed, this produce an n -cube since $\{o_i, i \in I\}$ commutes with each other. We need to record where the crossing changes take place and use something to mark them. This leads to the following definition.

An *n -singular knot* is an immersion of $K : \mathbb{S}^1 \rightarrow \mathbb{R}^3$ with n transversal double points. The set of n -singular up to isotopy is denoted by $\mathcal{K}_n$.

In particular, a 1-singular knot records how one knot turned into another by doing crossing changes. Intuitively, $\mathcal{K}_1$ are “tangent vectors” on the set of knots, and $\mathbb{Z}\mathcal{K}_1$ is “tangent bundle”. From the point of view of the previous section, n -singular knots are exactly n -cubes in the cubic complex of knots.

Given a singular knot K with a labeling set I of double points, for every $i \in I$ there are two ways to resolve the double point $\times$ labeled by i : the positive way $\nearrow$ and the negative way $\searrow$, which give rise to a well-defined resolution map

$$\begin{aligned} \partial_i : \mathbb{Z}\mathcal{K}_n &\rightarrow \mathbb{Z}\mathcal{K}_{n-1} \\ (\times) &\mapsto (\nearrow) - (\searrow) \end{aligned} \quad (2.6)$$

Inductively we have a complete resolution map $\prod_{i \in I} \partial_i : \mathbb{Z}\mathcal{K}_n \rightarrow \mathbb{Z}\mathcal{K}_0$. There are $n!$ choices of the order of this resolution, but the map is uniquely determined by:

$$\begin{aligned} \partial^n : \mathbb{Z}\mathcal{K}_n &\rightarrow \mathbb{Z}\mathcal{K}_0 \\ K &\mapsto \sum_{J \subset I} (-1)^{\#J} K_J \end{aligned} \quad (2.7)$$

where K_J is the knot obtained by substituting the double points labeled by J to the negative crossings and double points labeled by $I \setminus J$ by positive crossings.

For a knot invariant V , we can now define its n -th derivative by the pull-back $V^{(n)} := V \partial^n$ which is an invariant of n -singular knots. In the sequel we sometimes say that V is extended to an invariant of n -singular knot by the Vassiliev's skein relation

$$V^{(n+1)}(\times) = V^{(n)}(\nearrow) - V^{(n)}(\searrow). \quad (2.8)$$

For simplicity, we shall work over a base field $\mathbb{F}$ of characteristic 0. Let $\mathbb{F}\mathcal{K}_n$ be the $\mathbb{F}$ vector space by $\mathcal{K}_n$, then ∂^n naturally extends to $\mathbb{F}\mathcal{K}_n$. Denote $\mathcal{K}_n := \partial^n(\mathbb{F}\mathcal{K}_n)$, then

$$\mathcal{K}_0 \supset \mathcal{K}_1 \supset \cdots \supset \mathcal{K}_n \supset \cdots \quad (2.9)$$

is called the *singular knot filtration*. Its filtered dual is $\mathcal{V}_0 = \text{Hom}(\mathcal{K}_0, \mathbb{F})$ with ascending filtration

$$0 = \mathcal{V}_{-1} \subset \mathcal{V}_0 \subset \mathcal{V}_1 \subset \cdots \subset \mathcal{V}_n \subset \cdots \quad (2.10)$$

where $\mathcal{V}_n \subset \mathcal{V}_0$ consists of functionals that vanish on $\mathcal{K}_{n+1}$, or equivalently, its n -th derivative $V^{(n)} = V \partial^n = 0$. We call $V \in \mathcal{V}_n$ a *finite type invariant of degree n* . The union $\mathcal{V} = \bigcup_{n=0}^{\infty} \mathcal{V}_n$ is the space of finite type invariants.

Since finite type invariants can be thought of as polynomials in the space of invariants, it's natural to ask if we can use these invariants to approximate all knot invariants, just as the Stone-Weierstrass theorem does to continuous functions in real function space. An equally nontrivial question to ask is, does $\mathcal{V}$ distinguish knots? Or is $\mathcal{V}$ complete? The latter is equivalent to that the singular knot filtration is Hausdorff, i.e. $\mathcal{K}_{\infty} = \bigcap_n \mathcal{K}_n = 0$, or in other words $\mathcal{K} \rightarrow \widehat{\mathcal{K}} = \varprojlim \mathcal{K} / \mathcal{K}_n$ is injective. This conjecture is still open.

2.2.2 Classical Polynomials are Finite Type Invariants

It turns out most of our classical knot invariants are of finite type, or at least power series finite type invariants, namely, at least lie in $\widehat{\bigoplus_k V_k}$.

Proposition 2.2.1. *After substituting $t = e^x$, the n -th coefficient of Jones Polynomial is a finite type invariant of degree n .*

Proof. Assume it's true for $n \leq (k-1)$, so the evaluation of J on a k -singular knot only has terms of order $\geq k$. Note that the skein relation eq. (1.4) also holds for any singular knot, thus for a $(k+1)$ -singular knot, we have

$$\begin{aligned} J \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right) &= J \left(\begin{array}{c} \diagup \diagdown \\ \diagup \diagdown \end{array} \right) - J \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right) \\ &= (1 - e^{-x}) J \left(\begin{array}{c} \diagup \diagdown \\ \diagup \diagdown \end{array} \right) - (1 - e^x) J \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right) + (e^{x/2} - e^{-x/2}) J \left(\begin{array}{c} \diagup \diagdown \\ \diagup \diagdown \end{array} \right) \end{aligned} \quad (2.11)$$

it follows that $J \left(\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right)$ only has terms of order $\geq k+1$. $\square$

More generally, Birman and Lin proved in [7] that all quantum invariants are power series finite type invariants.

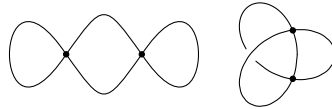
Proposition 2.2.2. *The n -th coefficient of Conway Polynomial is a finite type invariant of degree n .*

Proof. Conway's skein relation eq. (1.5) allows us to move the singular point with the effect multiplying t . Thus the value of C on any $(n+1)$ -singular knot only has terms of order $(n+1)$ or higher. $\square$

To analyze the size of $\mathcal{V}$, let's study it inductively.

Proposition 2.2.3. $\dim \mathcal{V}_0 = \dim \mathcal{V}_1 = 1, \dim \mathcal{V}_2 = 2$

Proof. Clearly $\mathcal{V}_0$ is just the space of constant functions since every knot can be unknotted after finitely many crossing changes. Let's look at $\mathcal{V}_1$, the functions that vanish on 2-singular knots. But again, by doing crossing changes, any 1-singular knot can be turned into a figure infinity knot: ∞ , hence the value of $V \in \mathcal{V}_1$ on singular 1-knot is constant, namely $\mathcal{V}_1 = \mathcal{V}_0$. But in the degree 2, by doing crossing changes and smooth deformations, we can reduce every 2-singular knot to one of the following:



(a) K_1 (b) K_2

Also, one cannot turn one of them into another. This is because if we go along K_1 , the singular points encountered are in order $abab$ while for K_2 it's $abba$, they differ even up to cyclic permutation. Thus, every function in $\mathcal{V}_2$ is determined by their value on K_1, K_2 , so $\dim \mathcal{V}_2 \leq 2$. But the Casson Invariant, i.e. the second coefficient of the Conway polynomial, is a non-constant invariant of degree 2 (for example $C(\bigcirc) = 1 + t^2$). Thus $\dim \mathcal{V}_2 = 2$. $\square$

Inductively, it's natural to ask the following question: How big is $\mathcal{V}_{n+1}/\mathcal{V}_n$? Is there a nice characterization of the associated graded space $\text{gr} \mathcal{V} = \bigoplus_n \mathcal{V}_n/\mathcal{V}_{n-1}$? Dually, we can study the linear space $\mathcal{K}$ spanned by knots and its associated graded space $\text{gr} \mathcal{K} = \bigoplus_n \mathcal{K}_n/\mathcal{K}_{n+1}$ directly. Note also there is a natural isomorphism $\mathcal{V}_{n-1}/\mathcal{V}_n \simeq (\mathcal{K}_n/\mathcal{K}_{n+1})^*$ thus $\text{gr} \mathcal{V}$ is the graded dual of $\text{gr} \mathcal{K}$. In particular, $\mathcal{V}_n \simeq (\mathcal{K}/\mathcal{K}_{n+1})^*$.

We shall answer these questions in the next few sections.

2.2.3 Bialgebra Structure

The space $\mathcal{K} = \mathbb{F}\mathcal{K}$ is a monoidal algebra, with commutative multiplication in $\mathcal{K}$ defined by connected sum of two knots (note that orientation is vital here), thus it has a natural commutative and cocommutative bialgebra structure $(\mathcal{K}, m, u, \delta, \epsilon)$ given by

$$\begin{aligned} m &: K_1 \otimes K_2 \mapsto K_1 \# K_2 \\ u &: 1 \mapsto \bigcirc \\ \delta &: K \mapsto K \otimes K \\ \epsilon &: K \mapsto 1 \end{aligned} \tag{2.12}$$

for knots K, K_1, K_2 . The singular knot filtration $(\mathcal{K}_n)$ is also compatible with this bialgebra structure:

$$m(\mathcal{K}_m \otimes \mathcal{K}_n) \subset \mathcal{K}_{m+n}, \delta \mathcal{K}_n \subset \sum_{p+q=n} \mathcal{K}_p \otimes \mathcal{K}_q. \tag{2.13}$$

The latter follows from the following lemma:

Lemma 2.2.4. *Write $K_J, J \subset I$ for the image under $\partial^{|J|}$ of the average formal sum of all possible complete desingularization of double points in $I \setminus J$ of K . Then*

$$\delta(\partial^n K) = \sum_J K_J \otimes K_{I \setminus J} \tag{2.14}$$

Proof. Assume this is true for $(n-1)$ -singular knot. For $K \in \mathcal{K}_n$, let K^+, K^- be the resolution at $i \in I$, let $I' = I \setminus i, J = J' \cup \{i\}$ then

$$\begin{aligned} \delta(\partial^n K) &= \delta(\partial^{n-1} K^+) - \delta(\partial^{n-1} K^-) \\ &= \sum_{J' \subset I'} (K_{J'}^+) \otimes (K_{I' \setminus J'}^+) - (K_{J'}^-) \otimes (K_{I' \setminus J'}^-) \\ &= \frac{1}{2} \sum_{J' \subset I'} (K_{J'}^+ - K_{J'}^-) \otimes (K_{I' \setminus J'}^+ + K_{I' \setminus J'}^-) + (K_{J'}^- + K_{J'}^+) \otimes (K_{I' \setminus J'}^+ - K_{I' \setminus J'}^-) \\ &= \sum_{J \subset I} K_J \otimes K_{I \setminus J} \end{aligned} \tag{2.15}$$

□

Dually, $\mathcal{V}$ has a filtered bialgebra structure $(\mathcal{V}, \delta^*, \epsilon^*, m^*, u^*)$ given by

$$\begin{aligned} \delta^*(V_1 \otimes V_2)(K) &= V_1(K) V_2(K) \\ \epsilon^* &: 1 \mapsto 1 \\ m^*(V)(K_1 \otimes K_2) &= V(K_1 \# K_2) \\ u^* &: V \mapsto V(\bigcirc) \end{aligned} \tag{2.16}$$

In particular the product of two invariants in $\mathcal{V}_m$ and $\mathcal{V}_n$ lies in $\mathcal{V}_{m+n}$, which is similar to polynomials. In fact we have the following Leibniz rule [30], which is the dual version of the formula eq. (2.14):

Proposition 2.2.5. *If K is a n -singular knot, V and W are invariants of the knot, then*

$$(V \cdot W)(\partial^n K) = \sum_{J \subset I} V(K_J) W(K_{I \setminus J}) \tag{2.17}$$

2.3 Chord Diagrams

In this section, we use some combinatorial methods to study finite type invariants or dually, the singular knot space.

2.3.1 Fundamental Theorem of Vassiliev Invariants

As we have noticed in the computation of $\mathcal{V}_2$, the ordered set of double points of a singular knot is unchanged up to cyclic permutations under crossing changes. It's convenient to use the following notation to encode this information graphically: A n -chord diagram on a compact oriented 1-manifold M (possibly with boundary) is the manifold M with an isotopy class of an embedding of an unordered set of n pairs of points into the interior of M . We shall use dashed chords to connect the points of a pair. For example, an 8-chord diagram on a manifold consists of 2 segments and a circle could look like:

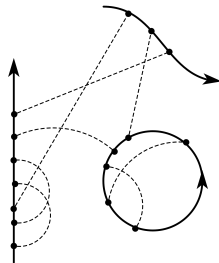


Figure 2.1: Chord Diagram of a Tangle

where chords are dashed and M is represented by solid lines. The $\mathbb{F}$ linear space spanned by n -chord diagrams on M is denoted by $\mathcal{D}_n(M)$. In this subsection we shall mainly consider the chord diagrams on $\mathbb{S}^1$, in this case, the circle is called the Wilson loop, and we simply write $\mathcal{D}_n$ for $\mathcal{D}_n(\mathbb{S}^1)$. For example: $\mathcal{D}_3$ is spanned by the following

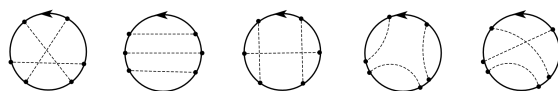


Figure 2.2: A basis of $\mathcal{D}_3$

Now given a n -singular knot $\mathbb{S}^1 \rightarrow \mathbb{R}^3$, we can draw its chord diagram by traveling along the circle and labeling the preimages of double points, then connect the preimage of the same double points with dashed chords. Conversely, from a chord diagram you can get a singular knot by first embedding the neighborhood of double points in $\mathbb{R}^3$, then joining them using disjoint arcs. If two singular knots $K, K' : \mathbb{S}^1 \rightarrow \mathbb{R}^3$ have the same chord diagrams, then they can be related by an isotopy of $\mathbb{R}^3$ that sends double points to the corresponding double points, then a homotopy of the arcs between them which can be decomposed into a series of isotopies and crossing changes. Thus there is a one-one correspondence:

$$\mathcal{K}_n / \langle \text{Crossing Changes} \rangle \iff n\text{-chord diagrams} \quad (2.18)$$

But since crossing changes in $\mathbb{FK}_n$ was mapped into $\mathcal{K}_{n+1}$ by ∂^n , it descends to

$$\mathcal{D}_n \simeq \mathbb{FK}_n / \langle \text{Crossing Changes} \rangle \xrightarrow{\partial^n} \mathcal{K}_n / \mathcal{K}_{n+1} \quad (2.19)$$

At this point, we can give a partial answer to the size of $\mathcal{V}_n / \mathcal{V}_{n-1} = (\mathcal{K}_n / \mathcal{K}_{n+1})^*$: it's bounded by $\mathcal{D}_n$, which is finite dimensional.

In fact, we can improve the bound. There are some obvious elements in $\mathbb{FK}_n$ that are mapped to 0 by ∂^n . For example, from a n -chord diagram with an isolated chord a (a chord that doesn't intersect other chords), one can correspond a n -singular knot with a double point a that divides the knot into two parts separated by a hyperplane across a . Clearly the resolution of this knot is 0. (fig. 2.3).

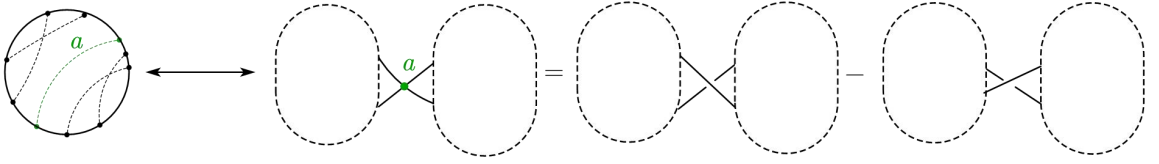


Figure 2.3: Topological 1T relation

“chord diagrams with isolated chords=0” is called 1T relation.

Another example is, note we have the following diagrams in fig. 2.4 for the same $(n-1)$ -singular knot,

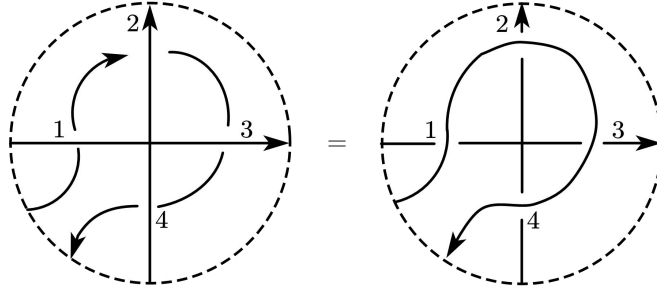
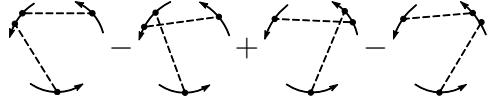


Figure 2.4: Topological 4T relation: two diagrams of a $(n-1)$ -singular knot

On the other hand, one can get from left to right in fig. 2.4 by flipping a sequence of crossings, labeled by 1,2,3,4. Let's write $K(\epsilon_i)_{i=1,\dots,4}$ for the knot whose crossing at i is a double point, an overcrossing and an undercrossing when $\epsilon_i = \cdot, +, -$. Then

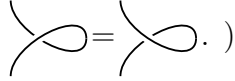
$$\begin{aligned} 0 &= \partial^{n-1} [K(+, -, -, +) - K(-, +, +, -)] \\ &= \partial^{n-1} [K(+, -, -, +) - K(-, -, -, +) + K(-, -, -, +) - K(-, +, -, +) \\ &\quad + K(-, +, -, +) - K(-, +, +, +) + K(-, +, +, +) - K(-, +, +, -)] \\ &= \partial^n [K(\bullet, -, -, +) - K(-, \bullet, -, +) - K(-, +, \bullet, +) + K(-, +, +, \bullet)] \end{aligned} \quad (2.20)$$

Thus eq. (2.19) vanishes on the following combination of chord diagrams



$$(2.21)$$

This is called the $4T$ relation. The parts of the 4 diagrams not shown in the figure should be the same. (In fact, since $4T$ relation allows us to move an isolated chord across the Wilson loop, the $1T$ relation can be weakened to: “diagrams with a chord relates two adjacent points=0” as in [23], which corresponds to the fact that the knot is unchanged under the first Reidemeister moves:

first Reidemeister moves: .)

Thus eq. (2.19) give rise to a map

$$\phi_n : \mathcal{D}_n / \langle 1T, 4T \rangle \rightarrow \mathcal{K}_n / \mathcal{K}_{n+1}, \quad (2.22)$$

which fits into a commutative diagram:

$$\begin{array}{ccc} \mathbb{FK}_n / \langle \text{Crossing changes} \rangle & \xlongequal{\quad} & \mathcal{D}_n \\ \partial^n \downarrow & & \downarrow \\ \mathcal{K}_n / \mathcal{K}_{n+1} & \xleftarrow{\phi_n} & \mathcal{D}_n / \langle 1T, 4T \rangle \end{array} \quad (2.23)$$

Let's denote $\mathcal{D}_n / \langle 1T, 4T \rangle$ by $\mathcal{A}_n$, and $\mathcal{D}_n / \langle 4T \rangle$ by $\mathcal{A}^c$, and let $\mathcal{A} = \bigoplus_n \mathcal{A}_n$. Then $(\phi_n)_{n \in \mathbb{N}}$ induce a surjective graded map

$$\phi : \mathcal{A} \rightarrow \text{gr } \mathcal{K} \quad (2.24)$$

Dually, we can consider functions on n -chord diagrams. A function of chord diagrams that vanishes on $1T, 4T$ relations can be viewed as an element in $\mathcal{W}_n = \mathcal{A}_n^*$, and is called a *homogeneous weight system of degree n* . The graded dual $\mathcal{W} := \mathcal{A}^* = \bigoplus \mathcal{W}_n$ of $\mathcal{A}$ is the space of all weight systems, which are functions on $\mathcal{A}$ that vanish on $\mathcal{A}_k, k \geq n$ for some n . Now we can relate finite type invariants with weight systems, dual to the previous process. The key observation is that the top degree derivative $V^{(n)}$ of a finite type invariant V of degree n is invariant under crossing changes on n -singular knots and hence depends only on the combinatorics of the parametrizing circle of n -singular knot, thus produce a functional on chord diagrams. But all such functionals are invariant under $1T$ and $4T$ relations, so every type n invariant actually produce a weight system via a map:

$$\text{Symb}_n : \mathcal{V}_n \rightarrow \mathcal{W}_n \quad (2.25)$$

($\text{Symb}_n(V)$ is called the “symbol” of V if V is of type n but not of type $n-1$ [14]). Since two invariants have the same n -th derivative if and only if they differ by a degree n invariant, Symb_n descends to an injective map $\phi_n^* : \mathcal{V}_n / \mathcal{V}_{n-1} \rightarrow \mathcal{W}_n$ which is literally the dual of ϕ_n

defined before. This process is illustrated in the following diagrams that dual to each other:

$$\begin{array}{ccccc}
\mathbb{F}\mathcal{K}_n & \longrightarrow & \mathbb{F}\mathcal{K}_n / \langle \text{crossing changes} \rangle & \xlongequal{\quad} & \mathcal{D}_n \\
\partial^n \downarrow & & \partial^n \downarrow & & \downarrow /1T, 4T \\
\mathcal{K} / \mathcal{K}_{n+1} & \longleftarrow & \mathcal{K}_n / \mathcal{K}_{n+1} & \xleftarrow{\phi_n} & \mathcal{A}_n \\
& & & & \\
\mathbb{F}\mathcal{K}_n^* & \longleftarrow & \mathbb{F}\mathcal{K}_n^* / \langle \text{crossing changes} \rangle^0 & \xlongequal{\quad} & \mathcal{D}_n^* \\
(-)^{(n)} \uparrow & & (-)^{(n)} \uparrow & & \uparrow \\
\mathcal{V}_n & \longrightarrow & \mathcal{V}_n / \mathcal{V}_{n-1} & \xrightarrow{\phi_n^*} & \mathcal{W}_n = \langle 1T, 4T \rangle^0
\end{array} \tag{2.26}$$

Note that Symb_{n+1} doesn't extend Symb_n , so we do not have a natural linear map $\mathcal{V} \rightarrow \mathcal{W}$ here, but ϕ_n^* induces an injective graded map

$$\phi^* : \text{gr}\mathcal{V} \rightarrow \mathcal{W} \tag{2.27}$$

Thus, we can characterize finite type invariants by weight systems, and the fundamental theorem of finite type invariants says they are essentially the same thing!

Let $\hat{\mathcal{A}}$ be the graded completion of $\mathcal{A}$ (This is the completion with respect to the associated descending filtration $\bigoplus_{k \geq n} \mathcal{A}_k$), as a vector space $\hat{\mathcal{A}} = \prod_n \mathcal{A}_n$. The fundamental theorem of finite type invariants states that ϕ and ϕ^* are isomorphisms with inverses given by a universal finite type invariant. More precisely:

Theorem 2.3.1 (Kontsevich[19]). *There exists a linear map $Z : \mathcal{K} \rightarrow \hat{\mathcal{A}}$ that is compatible the filtrations, namely*

$$Z(\mathcal{K}_n) \subset \prod_{k \geq n} \mathcal{A}_k \tag{2.28}$$

such that $(\text{gr}Z) \phi = \text{id}_{\mathcal{A}}$.

Unwinding the definitions, the theorem says we have a map $Z = \prod_{n=1}^{\infty} Z_n$ such that the n -th component Z_n is of type n , and for any n -singular knot K and the class D_K of its chord diagram in $\mathcal{A}_n$, we have $Z(\partial^n K) = Z^{(n)}(K) = D_K + \text{higher order terms}$.

Dually, the theorem can be rephrased as:

1. the filtered dual map of Z

$$Z^* : \mathcal{W} \xrightarrow{\sim} (\hat{\mathcal{A}})^* \xrightarrow{Z^*} \mathcal{K}^* = \mathcal{V} \tag{2.29}$$

where $(\hat{\mathcal{A}})^*$ is the filtered dual of $\hat{\mathcal{A}}$ with respect to the descending filtration $\prod_{k \geq n} \mathcal{A}_k$, is compatible with associated ascending filtrations.

2. the induced graded map $\text{gr}Z^* : \mathcal{W} = \text{gr}\mathcal{W} \rightarrow \text{gr}\mathcal{V}$ is the inverse of $\phi^* : \text{gr}\mathcal{V} \rightarrow \mathcal{W}$.

Unwinding these facts, we get the dual version of the fundamental theorem[4]:

Theorem 2.3.2. *We have linear an isomorphism $\text{gr}\mathcal{V} \cong \mathcal{W}$, more precisely, we have*

1. Any finite type invariant V of degree n produces a weight system $\text{Symb}(V)$ of degree n .
2. Any weight system W of degree n produces a finite type invariant $W(Z)$ of degree n .
3. $\text{Symb}(W(Z)) = W$ if W is of type n , and for any V of type n , $\text{Symb}(V)(Z)$ differs from V by a finite type invariant of degree $n - 1$.

Combined with proposition A.0.1, it follows that both $\widehat{Z} : \widehat{\mathcal{K}} \rightarrow \widehat{\mathcal{A}}$ and $Z^* : \mathcal{W} \rightarrow \mathcal{V}$ are isomorphisms. Note that although $\mathcal{V} \cong \text{gr}\mathcal{V}$ as vector spaces, this is not canonical. But Z gives a splitting of V via $(Z^*)^{-1}$. Such a map Z is called a universal finite type invariant, since any finite type invariant factors through it:

Corollary 2.3.3 (Universal Property). *A knot invariant V is of type n if and only if there is a linear functional W on $\widehat{\mathcal{A}}$ that vanishes on $\prod_{k \geq n+1} \mathcal{A}_k$, such that $V = W \circ Z$.*

Proof. This is a straightforward reformulation of the fact that the map Z^* is a filtration-respecting isomorphism. $\square$

The universal finite type invariant is not unique, since any filtration-respecting map $\mathcal{K} \rightarrow \widehat{\text{gr}\mathcal{K}}$ that induces identity $\text{gr}\mathcal{K} \rightarrow \text{gr}\mathcal{K}$ also gives rise to a universal finite type invariant. There are two famous constructions of universal finite type invariants, the Kontsevich integral Z_{KZ} that arises from Knizhnik-Zamoldchikov equations, and the configuration space integral that comes from the perturbative Chern-Simons theory Z_{CS} . It's conjectured that $Z_{KZ} = Z_{CS}$. We shall introduce them in the next two chapters.

2.3.2 Hopf Algebra Structure

There is a natural bialgebra structure on $\mathcal{A}, \mathcal{A}^c$ and $\mathcal{W}, \mathcal{W}^{fr}$, which is pulled back from $\text{gr}\mathcal{K}$ by ϕ . We focus on the unframed case, but the result holds for the framed case since the $1T$ relation is not essential here. The multiplication on $\mathcal{A}$ is just the connected sum of chord diagrams. One can easily verify this is independent of the cutting point by $4T$ relation. The unit is the diagram $\bigcirc$ with no chords.

The coalgebra structure on $\mathcal{A}$ is induced from eq. (2.14). Note that here we have a simpler form of δ on this graded level than on $\mathcal{K}$: if I is a labeling set of the chords, then

$$\delta(D) = \sum_{J \subset I} D_J \otimes D_{I \setminus J} \quad (2.30)$$

where D_J is the chord diagram consists of the chords labeled by J . This can be visualized as the sum of all possible splittings of D . For example:

$$\delta(\bigcirc \text{ with two vertical chords } \uparrow \downarrow) = \bigcirc \otimes \bigcirc \text{ with two vertical chords } \uparrow \downarrow + \bigcirc \text{ with two vertical chords } \uparrow \downarrow \otimes \bigcirc + \bigcirc \text{ with two vertical chords } \uparrow \downarrow \otimes \bigcirc + \bigcirc \text{ with two vertical chords } \uparrow \downarrow \otimes \bigcirc + 2 \bigcirc \text{ with one vertical chord } \uparrow \otimes \bigcirc \text{ with one vertical chord } \downarrow + 2 \bigcirc \text{ with one vertical chord } \downarrow \otimes \bigcirc \text{ with one vertical chord } \uparrow \quad (2.31)$$

We can also verify this is well-defined about $4T$ (or $1T$, which is trivial) relations, namely $\delta(\langle 4T \rangle) = \langle 4T \rangle \otimes \mathcal{A} + \mathcal{A} \otimes \langle 4T \rangle$. The counit takes value 1 on $\bigcirc$ and vanishes on all diagrams other than $\bigcirc$. It's straightforward to verify the compatibility of the structures. With these operations, $\phi : \mathcal{A} \rightarrow \text{gr}\mathcal{K}$ is a bialgebra epimorphism.

These combined indicate that $\mathcal{A}$ is a graded bialgebra. In fact, $\mathcal{A}$ is connected and of finite type, thus a Hopf algebra. Dually, the space of weight systems $\mathcal{W} = \mathcal{A}^*$ is also a

Hopf algebra, with the unit the function I_0 that takes value 1 on $\bigcirc$. Also note that the multiplication

$$(W_1 \cdot W_2)(D) = \sum_{J \subset I} W_1(D_J) W_2(D_{I \setminus J}) \quad (2.32)$$

of two weight systems W_1, W_2 is not the point-wise multiplication as functions on $\mathcal{A}$. Co-multiplication $m^*(W)(D_1 \otimes D_2) = W(D_1 \cdot D_2)$. The counit sends a weight system W to $W(\bigcirc)$. With these operations, $\phi^* : \text{gr}\mathcal{V} \rightarrow \mathcal{W}$ is a bialgebra monomorphism.

2.3.3 Horizontal Chord Diagrams

Another algebra of diagrams that will be used in the sequel is the algebra $\mathcal{A}^h$ of horizontal chord diagrams, these are chord diagrams on a 1-manifold $\sqcup_j I_j$ that consists of oriented intervals I_j .

A horizontal chord diagram on n strands with k chords is a chord diagram on n vertical intervals, oriented upwards along with k pairs of horizontal chords up to isotopy. Let $\mathcal{A}_k^h(n)$ be the space of horizontal chord diagrams on n strands with k chords, modulo the $4T$ relation, and let $\mathcal{A}^h(n) = \bigoplus_k \mathcal{A}_k^h(n)$ be the space of chord diagram with n strands. $\mathcal{A}^h(n)$ is an algebra with the multiplication defined by composition:

$$\begin{array}{c} \uparrow \uparrow \uparrow \uparrow \\ | \cdot | \cdot | \cdot | \\ \uparrow \uparrow \uparrow \uparrow \end{array} \cdot \begin{array}{c} \uparrow \uparrow \uparrow \uparrow \\ | \cdot | \cdot | \cdot | \\ \uparrow \uparrow \uparrow \uparrow \end{array} = \begin{array}{c} \uparrow \uparrow \uparrow \uparrow \\ | \cdot | \cdot | \cdot | \\ \uparrow \uparrow \uparrow \uparrow \end{array} \quad (2.33)$$

$\mathcal{A}^h(n)$ is generated by following diagrams of degree 1

$$D_{ij} = \begin{array}{c} \uparrow \uparrow \uparrow \uparrow \\ | \cdot | \cdot | \cdot | \\ \uparrow \uparrow \uparrow \uparrow \\ \text{...} \quad i \quad \text{...} \quad j \quad \text{...} \end{array} \quad (2.34)$$

subject to the horizontal $4T$ relation:

$$[D_{ij}, D_{ik} + D_{jk}] = 0, \quad \text{for distinct } i, j, k \quad (2.35)$$

and $2T$ relation:

$$[D_{ij}, D_{kl}] = 0, \quad \text{for distinct } i, j, k, l \quad (2.36)$$

These relations can be regarded as infinitesimal braid relation and far commutativity, thus the algebra of horizontal chord diagrams is the universal enveloping algebra of infinitesimal braid Lie algebra[18].

Note that horizontal chord diagrams are a special type of chord diagrams on a disjoint union of n intervals $\underbrace{\uparrow \cdots \uparrow}_n$, and horizontal $4T$ relation is the intersection of usual $4T$ relation

with $\mathcal{A}^h(n)$, hence there is a natural morphism $\mathcal{A}^h(n) \rightarrow \mathcal{A}^c \left(\underbrace{\uparrow \cdots \uparrow}_n \right)$. The injectivity of this morphism is nontrivial and was proved in [5].

2.4 Jacobi Diagrams: Bracket Arises

It's a little surprising, that knots, which are somewhat sophisticated and flexible, have some natural relations with Lie algebras, which are regular and rigid. But since we have restricted our attention to “Polynomial” invariants, this is not completely inconceivable.

This connection is done by extending the algebra $\mathcal{A}^c = \bigoplus_n \mathcal{D}_n / \langle 4T \rangle$ to an algebra $\mathcal{A}^t$ of diagrams involving trivalent vertices called Jacobi diagrams. A *Jacobi diagram* Γ on M is a finite uni-trivalent graph with at least 1 univalent vertex in each component, equipped with an isotopy class of embedding of univalent vertices $U \hookrightarrow M$, and an orientation on each trivalent vertex, namely a cyclic order (in this thesis, we use counter-clockwise order of the half-edges on the projection diagram to denote this orientation) on three half edges incident to the vertex. The degree of Γ is half the number of vertices. In particular, every chord diagram of degree n is a Jacobi diagram of degree n .

Let $\mathcal{D}^t(M)$ be the $\mathbb{F}$ -linear space spanned Jacobi diagrams of degree n , and $\mathcal{A}_n^t(M)$ be $\mathcal{D}_n^t(M)$ modulo the following *STU* relation:

$$\begin{array}{c} \text{Y-shaped diagram} \\ \text{S} \end{array} = \begin{array}{c} \text{Two parallel vertical lines} \\ \text{T} \end{array} - \begin{array}{c} \text{X-shaped diagram} \\ \text{U} \end{array} \quad (2.37)$$

As usual, the diagrams S, T, U are identical outside the fragment shown in the figure. Note that *STU* relation is homogeneous. Let $\mathcal{A}^t(M) = \bigoplus_n \mathcal{A}_n^t(M)$.

Since a chord diagram is a special type of Jacobi diagram, there is a degree preserving map $\mathcal{D}(M) \rightarrow \mathcal{A}^t(M)$. Also note that *STU* relation implies *4T* relation

$$\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} - \begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} = \begin{array}{c} \text{Diagram 5} \\ \text{Diagram 6} \end{array} = \begin{array}{c} \text{Diagram 7} \\ \text{Diagram 8} \end{array} - \begin{array}{c} \text{Diagram 9} \\ \text{Diagram 10} \end{array} \quad (2.38)$$

hence we have a well-defined map $\mathcal{A}^c(M) \rightarrow \mathcal{A}^t(M)$. It's easy to see that this map is surjective, since we can use *STU* relation to resolve the trivalent vertices and express any Jacobi diagram with a combination of chord diagrams. Surprisingly, this is an isomorphism:

Theorem 2.4.1 (Bracket-Rise). $\mathcal{A}^c(M) \cong \mathcal{A}^t(M)$

Proof. Although we have seen that $\rho : \mathcal{A}_n^c(M) \rightarrow \mathcal{A}_n^t(M)$ is surjective since every Jacobi diagram can be presented by a combination of chord diagrams, this expression might depend on the specific process to eliminate trivalent vertices and hence not unique. We shall show that the result of resolution is well-defined modulo *4T* relation. The proof is by building a map $\alpha : \mathcal{A}_n^t(M) \rightarrow \mathcal{A}_n^c(M)$ such that $\alpha\rho = \text{id}$ step by step. Let $\mathcal{D}_{n,k}^t$ be the subspace of $\mathcal{D}_n^t$ spanned by chord diagrams with k trivalent vertices, and let $\langle STU \rangle_k = \langle STU \rangle \cap (\mathcal{D}_{n,k-1}^t(M) \oplus \mathcal{D}_{n,k}^t(M))$. For $k = 0$, we set α_0 that takes every chord diagram to itself as a Jacobi diagram. *STU* relation in $\mathcal{D}_{n,0}^t$ is empty. Suppose we have defined for $i = 1, \dots, k-1$ a map

$$\alpha_i : \bigoplus_{j=1}^i \mathcal{D}_{n,j}^t(M) \rightarrow \mathcal{A}_n^c(M) \quad (2.39)$$

such that α_i extends α_{i-1} , and

$$\alpha_i(\langle STU \rangle_i) = 0 \quad (2.40)$$

This is equivalent to $\alpha_i(\text{Y}) = \alpha_{i-1}(\text{|||} - \text{X})$ at every edge that incident to a univalent vertex. For $D \in \mathcal{D}_{n,k}$, define $\alpha_k(D) = F(D, e)$ by resolving a selected edge e incident to a univalent vertex using the STU relation:

$$F(D = (\text{Y}), e) = \alpha_{k-1}(\text{|||} - \text{X}) \quad (2.41)$$

We must show that this also satisfies eq. (2.40) (or equivalently, $F(D, e)$ do not rely on the choice of the e). Let's prove this case by case:

1. The STU relation at an edge f that is disjoint from e is an immediate consequence of the fact that α_{k-1} satisfies eq. (2.40):

$$\begin{aligned} F\left(\left(\text{Y}_e \text{Y}_f\right), e\right) &= \alpha_{k-1}(\text{|||} \text{Y}) - \alpha_{k-1}(\text{X} \text{Y}) \\ &= \alpha_{k-2}(\text{|||} \text{|||}) - \alpha_{k-2}(\text{|||} \text{X}) \\ &\quad - \alpha_{k-2}(\text{X} \text{|||}) + \alpha_{k-2}(\text{|||} \text{X}) \\ &= \alpha_{k-1}(\text{Y} \text{|||}) - \alpha_{k-1}(\text{Y} \text{X}) \\ &= F\left(\left(\text{Y}_e \text{Y}_f\right), f\right) \end{aligned} \quad (2.42)$$

2. If e, f do intersect, and if there is an edge g that disjoint from e, f and incident to a univalent vertex, then by eq. (2.42), $F(D, e) = F(D, g) = F(D, f)$.
3. e, f, g join M with the same trivalent vertex. This case is immediate by $4T$ relation.
4. The last case, there is no such g , namely there are only 2 univalent vertices. In this case $F(D, e) = 0 = F(D, f)$.

$$F\left(\left(\text{...}\right), e\right) = \alpha_{k-1}\left(\text{...}\right) - \alpha_{k-1}\left(\text{...}\right) \quad (2.43)$$

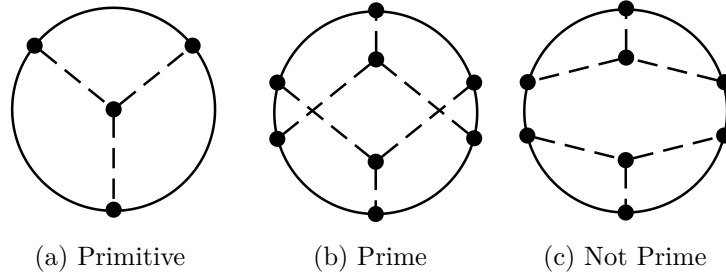
Note that we can expand ... as a sum of chord diagrams, and move the other chord across these chords using $4T$ relation, thus $F\left(\left(\text{...}\right), e\right) = 0$. Similarly

$$F\left(\left(\text{...}\right), f\right) := \alpha_{k-1}\left(\text{...}\right) - \alpha_{k-1}\left(\text{...}\right) = 0 \quad (2.44)$$

Inductively, we have built a map $\alpha : \bigoplus_{j=1}^{\infty} \mathcal{D}_{n,j}^t(M) \rightarrow \mathcal{A}_n^c(M)$, which takes all STU relations to 0, thus α desends to a well-defined map $\alpha : \mathcal{A}_n^t(M) \rightarrow \mathcal{A}_n^c(M)$ and $\alpha\rho = \text{id}$. This implies ρ is injective. $\square$

Due to the bracket-rise theorem, we shall write both spaces as $\mathcal{A}^{fr}(M)$, and the space $\mathcal{A}(M) = \mathcal{A}^{fr}(M) / \langle 1T \rangle$. As usual, we write $\mathcal{A}^{fr} = \bigoplus_n \mathcal{A}_n^{fr}$ for $\mathcal{A}^{fr}(\mathbb{S}^1) = \bigoplus_n \mathcal{A}_n^{fr}(\mathbb{S}^1)$. In this case, the space $\mathcal{A}^c$ is also endowed with a natural Hopf algebra structure (for general M the space $\mathcal{A}^{fr}(M)$ is only a coalgebra). The multiplication is defined by the connected sum and the co-multiplication is defined by the sum of all possible splittings of diagrams. Since these operations are well-defined modulo the STU relation and they agree with operations on $\mathcal{A}^c$, we can say $\mathcal{A}^c \cong \mathcal{A}^t$ as Hopf algebras. Also, it's easy to see that Jacobi diagrams whose graph is connected are primitive. In fact, by virtue of the Milnor-Moore theorem [A.0.2](#), the primitive subspace is spanned by connected Jacobi diagrams [\[4\]](#), and every diagram can be uniquely expressed as a polynomial of primitive ones. As a consequence:

Lemma 2.4.2. *there is a unique projection map onto the primitive subspace $P : \mathcal{A} \rightarrow \mathcal{P}(\mathcal{A})$ that vanishes on the non-prime graphs. A primitive diagram is prime (cannot be expressed as a multiplication of two diagrams).*



The only group-like element in $\mathcal{W}$ is the unit I_0 : the function that takes the value 1 on $\bigcirc$ and vanishes on other diagrams. The primitive weight systems are those W satisfying $W(D_1 \cdot D_2) = W(D_1)I_0(D_2) + I_0(D_1)W(D_2)$ for $D_1, D_2 \in \mathcal{A}$, so they vanish on any non-prime diagrams, so W is primitive if and only if it factors through P . The Milnor-Moore theorem [A.0.2](#) shows that any weight system can be uniquely expressed as a polynomial of primitive weight systems.

Now let's briefly explain how Lie algebra arises:

Lemma 2.4.3. *STU relation implies the following antisymmetry (AS) relation*

$$\begin{array}{c} \diagup \\ \diagdown \end{array} = - \begin{array}{c} \diagdown \\ \diagup \end{array} \quad (2.45)$$

and IHX relation

$$\begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} = \begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} - \begin{array}{c} \diagup \bullet \diagdown \\ \diagdown \bullet \diagup \end{array} \quad (2.46)$$

I
 H
 X

of internal vertices.

Proof. Use induction on the distance of the vertex to M , we prove the distance 1 case, then we can use STU relation to move the vertex closer to M by eliminating an edge incident to M that lies on the shortest path from the vertex to M .

AS relation on vertices has distance 1 with the manifold is clear by STU relation:

$$\begin{array}{c} \diagup \quad \diagdown \\ \bullet \\ \hline \end{array} = \begin{array}{c} \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \hline \end{array} - \begin{array}{c} \diagdown \quad \diagup \\ \bullet \quad \bullet \\ \hline \end{array} = \begin{array}{c} \diagup \quad \diagdown \\ \bullet \\ \hline \end{array} \quad (2.47)$$

The IHX relation for distance 1 case is proved by using STU relations twice (Note that the particular edge that is adjacent to M doesn't matter since IHX relation is preserved after a rotation of 90 degree, and here we take the bottom right edges connects with M)

$$\begin{aligned} I &= \begin{array}{c} b \\ \diagup \quad \diagdown \\ \bullet \\ \hline \end{array} c = (abc) - (acb) - (bca) + (cba) \\ H &= \begin{array}{c} b \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \hline \end{array} c = (abc) - (bac) - (cab) + (cba) \\ X &= \begin{array}{c} b \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \\ \hline \end{array} c = (acb) - (cab) - (bac) + (bca) \end{aligned} \quad (2.48)$$

where $(\sigma(a)\sigma(b)\sigma(c))$ is the diagram such that when moving along M , the 3 points on M is connected by a chord to $\sigma(a), \sigma(b), \sigma(c)$ respectively. clearly $I = H - X$ in this case. $\square$

Let's fix a Lie algebra $\mathfrak{g}$, namely a vector space equipped with a bilinear map $[-, -] \in \mathfrak{g}^* \otimes \mathfrak{g}^* \otimes \mathfrak{g}$, subject to antisymmetry property

$$[a, b] = -[b, a] \quad (2.49)$$

and Jacobi identity

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0 \quad (2.50)$$

And in addition we choose a nondegenerate $\mathfrak{g}$ invariant symmetric bilinear form $\langle -, - \rangle \in \mathfrak{g}^* \otimes \mathfrak{g}^*$ (e.g. the Cartan Killing form of semi-simple $\mathfrak{g}$) so that we have a canonical isomorphism $\mathfrak{g} \cong \mathfrak{g}^*$ and $[-, -] \in \mathfrak{g}^* \otimes \mathfrak{g}^* \otimes \mathfrak{g}^*$. We also choose a representation V of $\mathfrak{g}$, namely a map in $\mathfrak{g} \otimes V^* \otimes V$ such that for $u \in V, a, b \in \mathfrak{g}$:

$$([a, b])u = a(bu) - b(au) \quad (2.51)$$

If we look at eq. (2.45), and think of the trivalent vertices as the bracket, then AS relation eq. (2.45) is exactly anti-symmetry property of the bracket. If we examine the IHX relation, then it's exactly the Jacobi identity of the bracket. Now if we think of a segment of the manifold as $V^* \otimes V$, and interpret the univalent vertex as $\mathfrak{g} \otimes V^* \otimes V$ then STU relation is nothing but the definition eq. (2.51) of the representation! Now, concatenating two diagrams can be viewed as the contraction of two tensors. In particular, if $M = \mathbb{S}^1$, a Jacobi diagram corresponds to an element in the base field $\mathbb{F}$. In summary, we have built a function on $\mathcal{A}(\mathbb{S}^1)$, namely a framed weight system.

This observation can be made precise as follows:

Theorem 2.4.4 (Bar-Natan[4]). *Given a finite dimensional Lie algebra $\mathfrak{g}$, an invariant symmetric non-degenerate bilinear form $\langle -, - \rangle$, and a finite dimensional representation U , there is a natural way to construct a weight system of degree n :*

$$W_{\mathfrak{g}, U, n} : \mathcal{A}_n \rightarrow \mathbb{F} \tag{2.52}$$

Chapter 3

The Kontsevich Integral

3.1 Knizhnik-Zamolodchikov Equation

3.1.1 Iterated Integral

The theory of iterated integrals was invented by K.T. Chen[13] to study homotopy theory using a de Rham theoretical approach, it turns out that iterated integral can be used to explicitly express the monodromy representation of connections. Based on Chen's work, Kohno [18] is able to study finite type invariants for braids. The Kontsevich integral is a natural (but nontrivial) generalization to links.

In this thesis, given 1-forms $\omega_1(t_1), \dots, \omega_n(t_n)$ on the interval $[0, 1]$, we shall follow Chen's notation to write

$$\begin{aligned} \int_0^1 \bigwedge_{k=1}^n \omega_k(t_k) &:= \int_0^1 \omega_1(t_1) \cdots \omega_n(t_n) := \int_{\Delta^n} p_1^* \omega_1(t_1) \wedge \cdots \wedge p_n^* \omega_n(t_n) \\ &= \int_0^1 \omega_1(t_1) \left(\cdots \left(\int_0^{t_{n-1}} \omega_n(t_n) \right) \cdots \right) \end{aligned} \quad (3.1)$$

where $\Delta^n = \{0 \leq t_n \leq \cdots \leq t_1 \leq 1\}$ and $p_j : \Delta^n \rightarrow [0, 1]$ is the i -th projection.

Example 3.1.1. Iterated integrals in calculus have the form

$$I = \int_0^1 f_1(t_1) dt_1 \cdots f_n(t_n) dt_n := \int_{0 \leq t_n \leq \cdots \leq t_1 \leq 1} f_1(t_1) \cdots f_n(t_n) dt_n \cdots dt_1 \quad (3.2)$$

For example take

$$f_n(t) = \frac{1}{1-t}, f_1(t) = \cdots = f_{n-1}(t) = \frac{1}{t} \quad (3.3)$$

then

$$\begin{aligned} I &= \int_0^1 \frac{dt_1}{t_1} \cdots \frac{dt_{n-1}}{t_{n-1}} (-\log(1-t_{n-1})) dt_{n-1} \\ &= \sum_{k=1}^{\infty} \int_0^1 \frac{dt_1}{t_1} \cdots \frac{dt_{n-2}}{t_{n-2}} \frac{d(t_{n-1})^k}{k^2} \\ &= \sum_{k=1}^{\infty} \frac{1}{k^n} \\ &= \zeta(n) \end{aligned} \quad (3.4)$$

Here we can change the order of integration and summation because $\sum x^k/k^{n+1}$ converges uniformly when $n \geq 1$.

Iterated integrals arise naturally in solving equations of the form

$$\frac{dx}{dt} = Ax \quad (3.5)$$

The simplest example is the Picard iteration in the proof of Picard-Lindelöf theorem for ODE. Let $A(t)$ be a $n \times n$ matrix, consider the equation for $x \in \mathbb{R}^n$

$$\frac{dx}{dt} = A(t)x, x(0) = c \quad (3.6)$$

which is equivalent to find $Y(t) \in \text{End}(\mathbb{R}^n), x(t) = cY(t)$ s.t.

$$\frac{dY(t)}{dt} = A(t)Y(t), Y(0) = I \quad (3.7)$$

We can solve this equation formally, by considering the equation of a function $Y_{\hbar}(t)$ that takes values in $\mathbb{C}[[\hbar]]$ where $\hbar$ is an intermediate:

$$\frac{dY_{\hbar}(t)}{dt} = \hbar A(t)Y_{\hbar}(t), Y(0) = I \quad (3.8)$$

and the perturbative expansion

$$Y_{\hbar}(t) = I + Y_1(t)\hbar + Y_2(t)\hbar^2 \dots \quad (3.9)$$

Then the equation eq. (3.8) is equivalent to a system of equations

$$\frac{d}{dt}Y_j(t) = A(t)Y_{j-1}(t) \quad (3.10)$$

We can solve these equations by integrate $Y_j(t)$ iteratively. A solution can be written as

$$\begin{aligned} Y_{\hbar}(t) &= I + \hbar \int_0^t A(t_1)dt_1 + \hbar^2 \int_0^t A(t_1) \int_0^{t_1} A(t_2)dt_2dt_1 + \dots \\ &= \sum_{m=0}^{\infty} \hbar^m \int_0^t A(t_1)dt_1 \dots A(t_m)dt_m \end{aligned} \quad (3.11)$$

Setting $\hbar = 1$ would give us the original Picard iteration.

We can generalize eq. (3.8) and eq. (3.9) to a path with values in a completed graded algebra A with the linear topology determined by its associated descending filtration (which agrees with the product topology), following Chen's formal differential equation theory in [12]. Consider a path $Y : [0, 1] \rightarrow A$, the derivative is defined by $\lim_{h \rightarrow 0} (Y(t+h) - Y(t))/h$. Let $A(t) \subset A_1$ be a path into the homogeneous component of degree 1, then $dY(t)/dt = A(t)Y(t)$ can be solved (under some hypothesis about the topology of A) iteratively as eq. (3.11).

3.1.2 Flat Connection and Parallel Transport

Let's move on to a manifold M , with A an algebra over $\mathbb{C}$, and V a finite-dimensional representation of A . We shall consider the equation of an V -valued function $X \in \Gamma(M \times V)$

$$dX = \omega \cdot X \quad (3.12)$$

where $\omega \in A \otimes \Omega^1(M)$ is an A -valued 1-form. In the context of differential geometry, $\nabla = d - \omega$ is an A -valued connection on the trivial bundle $M \times V$. Fixing a global trivialization, $(-\omega)$ is the corresponding connection form, a solution of $\nabla X = 0$ is a flat section. System eq. (3.12) is over-determined, so not always solvable. To make sure Equation (3.12) has a solution for every initial condition, $\nabla^2 X = (\omega \wedge \omega - d\omega)X = 0$, so ∇ must be flat. One can utilize the Frobenius theorem on the bundle $M \times V$ to show that this is sufficient. Here we use an elementary and direct method to prove the existence theorem.

We can express the parallel section by parallel transport along a curve $\gamma : [0, 1] \rightarrow M$ with endpoints $\gamma(0) = x_0$ and $\gamma(1) = x$. We can solve the equation eq. (3.12) by first solving the equation along the path γ :

$$\frac{df}{dt} = \omega(\dot{\gamma}) \cdot f, f(0) = v_0 \quad (3.13)$$

Let f_γ be the solution of eq. (3.13). It's well-known that a connection is flat if and only if the parallel transport is homotopy invariant, and if and only if it admits a basis of flat sections. Here we give a direct proof (which essentially treats a special case of Frobenius theorem):

Theorem 3.1.2 (Monodromy Theorem). *The solution f_γ of eq. (3.13) is unchanged after a fix-end-point homotopy of γ under the integrability condition*

$$d\omega - \omega \wedge \omega = 0 \quad (3.14)$$

Proof. It suffices to prove the theorem locally. Consider a homotopy $h : [0, 1] \times [0, 1] \rightarrow U \subset M$ in a neighborhood U of x_0 that sends $\gamma = h_0 = h(-, 0)$ to $h_1 = h(-, 1)$. Suppose $h^*(\omega) = p(s, t)ds + q(s, t)dt$. Then eq. (3.14) can be expressed as

$$-\frac{\partial q}{\partial s} + \frac{\partial p}{\partial t} + [p, q] = 0 \quad (3.15)$$

Let $P(s, t), Q(s, t)$ be solutions of the following

$$\begin{aligned} \frac{\partial P}{\partial t} &= pP, P(0, t) = I \\ \frac{\partial Q}{\partial s} &= qQ, Q(s, 0) = I \end{aligned} \quad (3.16)$$

shrink the neighbourhood U so that Q is invertible. Then $P(s, t)v$ is the parallel transport of v from $h(0, t)$ to $h(s, t)$ along the curve $h(-, t)$, and $Q(s, t)v$ gives the parallel transport of v from $h(s, 0)$ to $h(s, t)$ along the curve $h(s, -)$. Note that

$$f(s, t) = \frac{\partial Q}{\partial s} - pQ + Q \cdot p(s, 0) \quad (3.17)$$

satisfies

$$\frac{\partial f}{\partial t} = qf, f(s, 0) = 0 \quad (3.18)$$

thus $f \equiv 0$ and

$$f(s, t) = \frac{\partial Q}{\partial s} - pQ + Q \cdot p(s, 0) \quad (3.19)$$

Consider the following path h_L in fig. 3.1

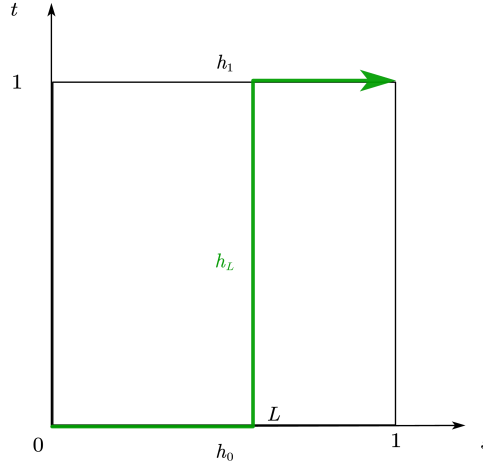


Figure 3.1: Homotopy Invariance

then the result of parallel transport is $f_{h_L}(1) = P(1, 1)P(L, 1)^{-1}Q(L, 1)P(L, 0)$ and use that

$$\frac{\partial P}{\partial s} = -P^{-1} \frac{\partial P}{\partial s} \quad (3.20)$$

Combine this with eq. (3.19)

$$\frac{\partial f_{h_T}(1)}{\partial T} = 0 \quad (3.21)$$

It follows that $f_{h_0}(1) = f_{h_1}(1)$. □

By virtue of the monodromy theorem, we immediately get

Theorem 3.1.3 (Local Integrability). *The total differential equation eq. (3.12) has a unique local solution near a point $x_0 \in M$ given an initial condition $X(x_0) = v_0 \in V$ under the condition of eq. (3.13)*

One should not expect that a global solution exists, as there are multiple ways to extend the function X along different paths by gluing together local solutions, just like in the case of analytic continuation in the complex analysis. In the special case of the loop, we get a monodromy representation $\pi_1(M) \rightarrow \text{End}(V)$.

3.1.3 Monodromy Representation of Braids and Regular String Links

n -braids, or more generally regular string links with n strands, can be regarded as paths in the configuration space $\text{Conf}_n(\mathbb{C})$ of n distinct points on the complex plane $\mathbb{C}$.

An interesting class of flat connections in this case comes from Lie algebras. Let $A = U(\mathfrak{g})$ be the universal enveloping algebra of a metric Lie algebra $\mathfrak{g}$ over $\mathbb{C}$. Choosing an orthogonal basis $\{I_k\}$ of $\mathfrak{g}$, then $C = \sum_k I_k \otimes I_k$ is the Casimir element determined by the metric, set

$$\tau_{ij} = \sum_k \underbrace{1 \otimes \cdots \otimes I_k}_{i} \otimes 1 \otimes \cdots \otimes 1 \otimes I_k \otimes \cdots \otimes 1 \in U(\mathfrak{g})^{\otimes n} \quad (3.22)$$

The most important property of τ_{ij} is:

Proposition 3.1.4. *For distinct i, j, k, l , there are $2T$ and $4T$ relations:*

$$[\tau_{ij}, \tau_{kl}] = 0, [\tau_{ij}, \tau_{ik} + \tau_{jk}] = 0 \quad (3.23)$$

The connection form we are considering is $U(\mathfrak{g})$ -valued one form

$$\omega = \sum_{1 \leq j < k \leq n} \tau_{jk} \omega_{jk}, \omega_{jk} := d \log(z_j - z_k) \quad (3.24)$$

Thus the equation eq. (3.12) becomes

$$dX = \frac{1}{2\pi\sqrt{-1}} \left(\sum_{1 \leq j < k \leq n} \tau_{jk} \omega_{jk} \right) \cdot X \quad (3.25)$$

which is called *Knizhnik-Zamolodchikov equation*.

Let's show it satisfies integrability condition eq. (3.14). This boils down to examine the condition

$$\omega \wedge \omega = \sum_{i < j, k < l} \tau_{ij} \tau_{kl} d \log(z_i - z_j) \wedge d \log(z_k - z_l) = 0. \quad (3.26)$$

In fact, we have:

Proposition 3.1.5. *For any τ_{ij}, ω_{ij} satisfies $2T$ and $4T$ relation eq. (3.23) and Anord's identity eq. (1.12), the connection*

$$\omega = \sum_{1 \leq i < j \leq n} \tau_{ij} \omega_{ij}$$

satisfies $\omega \wedge \omega = 0$.

Proof. The sum

$$\sum_{i < j, k < l} \tau_{ij} \tau_{kl} \omega_{ij} \wedge \omega_{kl} \quad (3.27)$$

splits into 3 parts according to $\# \{i, j, k, l\} \in \{2, 3, 4\}$. If $\# \{i, j, k, l\} = 2$, $\omega_{ij} = \omega_{kl}$ so their wedge product vanishes. If $\# \{i, j, k, l\} = 4$, then τ_{ij} commutes with τ_{kl} by $2T$ relation but

$[\omega_{ij} \text{ and } \omega_{kl}] = 0$ anti-commutative, again the terms cancel each other. If $\#\{i, j, k, l\} = 3$, then sum equals

$$\begin{aligned} & \sum_{i < j < k} [\tau_{ij}, \tau_{jk}] \omega_{ij} \wedge \omega_{jk} + [\tau_{jk}, \tau_{ik}] \omega_{jk} \wedge \omega_{ki} + [\tau_{ki}, \tau_{ij}] \omega_{ki} \wedge \omega_{ij} \\ &= \sum_{i < j < k} [\tau_{ij}, \tau_{jk}] (\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij}) \\ &= 0. \end{aligned} \tag{3.28}$$

□

Thus, the Knizhnik-Zamolodchikov connection is flat, and the parallel transport map

$$P_\gamma : V^{\otimes n} \rightarrow V^{\otimes n}, f_\gamma(0) \mapsto f_\gamma(1) \tag{3.29}$$

is homotopy invariant of γ and hence give rise to a representation of fundamental groupoid:

$$\begin{aligned} P : \Pi_1(\text{Conf}_n(\mathbb{C})) &\rightarrow \text{End}(V^{\otimes n}) \\ [\gamma] &\mapsto P_\gamma \end{aligned} \tag{3.30}$$

Restrict this on loops, we get a monodromy representation of the pure braid group:

$$P_n \rightarrow \text{End}(V^{\otimes n}) \tag{3.31}$$

factor out the action of symmetry group S_n , we have a monodromy representation of braid group:

$$B_n \rightarrow \text{End}(V^{\otimes n}) \tag{3.32}$$

Similar to the discussion in section 3.1.1, this monodromy representation can be presented explicitly by an iterated integral

$$P_\gamma = \sum_{n=0}^{\infty} \int_0^1 \omega(t_1) \cdots \omega(t_n) \tag{3.33}$$

where $\omega(t) = \omega(\dot{\gamma})dt$ is the pull back of the KZ connection ω by γ .

3.1.4 Formal Knizhnik-Zamolodchikov Equation

In a similar manner to our approach to solving linear ordinary differential equations in section 3.1.1, let's consider a formal equation based on our previous discussion. In some sense, this equation is a universal Knizhnik-Zamolodchikov equation. We shall replace Lie algebra $\mathfrak{g}$ and its representation V with Jacobi diagrams, which can be regarded as universal among all $(\mathfrak{g}, V)$.

Formal Knizhnik-Zamolodchikov equation is an equation for a $\widehat{\mathcal{A}^h}$ -valued function $X : \text{Conf}_n(\mathbb{C}) \rightarrow \widehat{\mathcal{A}^h}$,

$$dX = \frac{1}{2\pi\sqrt{-1}} \sum_{1 \leq i < j \leq n} D_{ij} d\log(z_i - z_j) X \tag{3.34}$$

By eq. (2.35), eq. (2.36), eq. (1.12) and proposition 3.1.5 the formal KZ connection form

$$\omega_{KZ} = \frac{1}{2\pi\sqrt{-1}} \sum_{1 \leq i < j \leq n} D_{ij} d \log(z_i - z_j) \quad (3.35)$$

is flat. Thus, the transport P_γ is fix-end-point homotopy invariant, and it can be presented by eq. (3.33). Let's compute this formula explicitly:

$$\begin{aligned} P_\gamma &= \sum_{m=0}^{\infty} \int_0^1 \bigwedge_{k=1}^m \left(\frac{1}{2\pi\sqrt{-1}} \sum_{1 \leq i < j \leq n} D_{ij} d \log(z_i(t_k) - z_j(t_k)) \right) \\ &= \sum_{m=0}^{\infty} \left(\frac{1}{2\pi\sqrt{-1}} \right)^m \int_0^1 \sum_{1 \leq i_k < j_k \leq m} \left(\prod_{k=1}^m D_{i_k j_k} \right) \left(\bigwedge_{k=1}^m d \log(z_{i_k}(t_k) - z_{j_k}(t_k)) \right) \\ &= \sum_{m=0}^{\infty} \left(\frac{1}{2\pi\sqrt{-1}} \right)^m \int_0^1 \sum_P D_P \bigwedge_{k=1}^m d \log(z_{i_k}(t_k) - z_{j_k}(t_k)) \end{aligned} \quad (3.36)$$

where P is a function $\{1, \dots, m\} \rightarrow \{1, \dots, n\} \times \{1, \dots, n\}, k \mapsto (i_k, j_k), i_k < j_k$, which can be regarded as a configuration of horizontal chords, and D_P is the corresponding horizontal chord diagram in $\mathcal{A}_m^h(n)$.

To deal with tangles, we need a bigger algebra than $\mathcal{A}^h(n)$. The connection eq. (3.35) need to be generalized to $\mathcal{A}(\underbrace{\uparrow \dots \uparrow}_m \underbrace{\downarrow \dots \downarrow}_n)$ (note that here we have factor out the $1T$ relation)

$$\omega_{m,n} = \frac{1}{2\pi\sqrt{-1}} \sum_{1 \leq i < j \leq m+n} s_i s_j D_{ij} d \log(z_i - z_j) \quad (3.37)$$

where

$$s_i = \begin{cases} 1 & \text{if } i \leq m \\ -1 & \text{if } i > m \end{cases} \quad (3.38)$$

and D_{ij} is a diagram with just one chord connecting the i -th and j -th vertical lines. Again we have $2T$ relation

$$[s_i s_j D_{ij}, s_k s_l D_{kl}] = 0 \quad \text{for distinct } i, j, k, l \quad (3.39)$$

The $4T$ relation implies a similar horizontal $4T$ relation, taking into account the signs

$$[s_i s_j D_{ij}, s_i s_k D_{ik} + s_j s_k D_{j,k}] = 0 \quad \text{for distinct } i, j, k \quad (3.40)$$

Combining these with eq. (1.12), this connection is also flat by proposition 3.1.5. Thus, we can explicitly compute the transport along γ as in eq. (3.36):

$$P_\gamma = \sum_0^{\infty} \left(\frac{1}{2\pi\sqrt{-1}} \right)^m \int_0^1 \sum_P (-1)^{\downarrow_P} D_P \bigwedge_{k=1}^m d \log(z_{i_k}(t_k) - z_{j_k}(t_k)) \quad (3.41)$$

where $\downarrow_P$ is the number of points in a pair in P that is oriented downwards. This integral is a prototype of the Kontsevich integral for arbitrary tangles.

3.2 The Kontsevich Integral

3.2.1 Analytic Construction of the Kontsevich Integral for Tangles

To streamline the theory, I shall start by introducing the Kontsevich integrals for arbitrary strict Morse tangle. A tangle is an embedding $T : M \rightarrow \mathbb{C} \times [0, 1]$ of a compact manifold M , so that $T(\partial M) \subset \mathbb{R} \subset \mathbb{C} \times \{0, 1\}$, considered up to isotopy. $T^{-1}(\mathbb{R} \times \{1\})$ is called the top of T , and $T^{-1}(\mathbb{R} \times \{0\})$ is the bottom. We say T is regular if $t : \mathbb{C} \times [0, 1] \rightarrow [0, 1]$ has no critical points on $T(M)$. Let $\mathcal{T}$ be the $\mathbb{F}$ -linear space of tangles with support M . There is a tensor product $T_1 \otimes T_2$ for any tangles $T_1 \in \mathcal{T}(M_1), T_2 \in \mathcal{T}(M_2)$, by placing them adjacent to each other. There is also an obvious product by stacking T_1 above T_2 when the top of T_2 coincides with the bottom of T_1 .

As we have discussed in the previous section, iterated integral is invariant under homotopy relative to the endpoints of a curve in configuration spaces, in particular, they give invariants of braids and regular string links. Now we generalize this to tangles. Fix a compact 1-manifold M and let $\mathcal{T}(M)$ be the linear space of tangles on M . Consider a decomposition $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}_t$ and choose tangle embeddings $T : M \rightarrow \mathbb{C} \times [0, 1]$ such that the restriction of $t : \mathbb{R}^3 \rightarrow \mathbb{R}_t$ on T is a strict Morse function. The Kontsevich integral is defined as a map

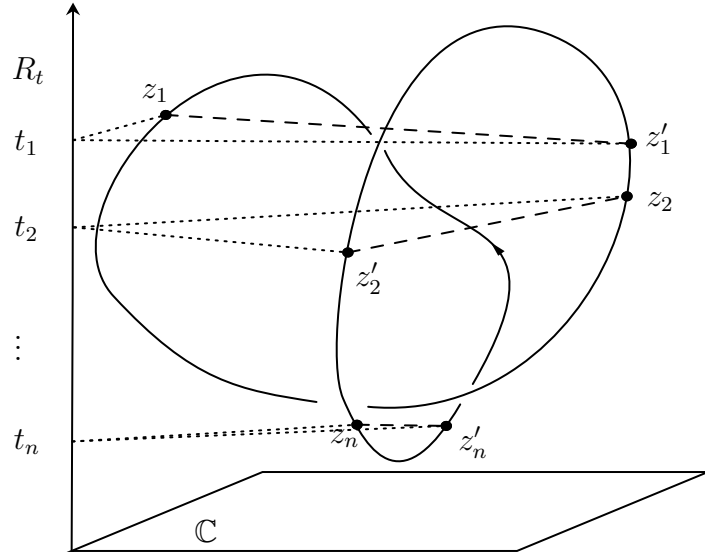


Figure 3.2: A configuration P on the unknot

that takes values in the graded complement $\widehat{\mathcal{A}(M)}$ of the tangle chord diagrams on M over $\mathbb{C}$, modulo $1T$ and $4T$ relations:

$$I(T) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi\sqrt{-1}} \right)^n \int_{0 < t_n < \dots < t_1 < 1} \sum_P (-1)^{\downarrow_P} D_P \bigwedge_{j=1}^n d \log(z_j(t_j) - z'_j(t_j)) \quad (3.42)$$

where P is a configuration space of horizontal n -chords consists of n pairs

$$\left\{ ((z_j, t_j), (z'_j, t_j)) \in \text{Conf}_2(T(M)), j = 1, \dots, n, t_j > t_{j+1} \right\} \quad (3.43)$$

and $\downarrow_P$ is the number of points for a configuration in P that is oriented downwards (see section 3.2.1). D_P is the class of tangle chord diagram associated with P in $\widehat{\mathcal{A}(M)}$. For example, for P on the unknot in section 3.2.1 D_P looks like:

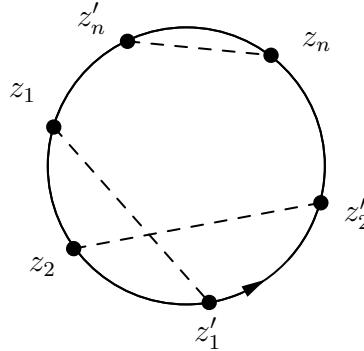


Figure 3.3: D_P

The sum runs over all possible configurations P . The form $d \log(z_j(t_j) - z'_j(t_j))$ is the pull back of $d \log(z_j - z'_j)$ on configuration space to the integral domain by the parametrization $t_j \mapsto (z_j, z'_j)$. As usual, the $n = 0$ term is the diagram $\bigcirc$ with no chords.

Let's see some examples:

Example 3.2.1. The linking number formula eq. (1.26) is exactly the coefficient of $\bigcirc \rightarrow \bigcirc$ in the Kontsevich integral of the link. We shouldn't expect to get any new invariant from its higher degree analogues, since these integrals are invariant under crossing changes inside one component, and the link modulo these crossing changes is uniquely determined by the linking number. For example, the coefficient of $\bigcirc \dashrightarrow \bigcirc$ is $(lk(L_1, L_2))^2/2$, and the sum of coefficients of diagrams with n chords connecting two link components is $(lk(L_1, L_2))^n/n!$.

Example 3.2.2. Let's compute the Kontsevich integral of the elementary braids $\overline{\sum}$ and $\underline{\sum}$. It's easy to see that each iteration is exactly the braiding number Wind_i of the strands in $[0, t_i]$, so the n -th coefficient is

$$\begin{aligned} \int_0^{\text{Wind}} d\text{Wind}_1 \cdots d\text{Wind}_n &= \int_0^{\text{Wind}} d\text{Wind}_1 \cdots \int_0^{\text{Wind}_{n-2}} d\text{Wind}_{n-1} \int_0^{\text{Wind}_{n-1}} d\text{Wind}_n \\ &= \frac{(\text{Wind})^n}{n!} \end{aligned} \quad (3.44)$$

For $\overline{\sum}$, $\text{Wind} = 1/2$, thus

$$\begin{aligned} I\left(\overline{\sum}\right) &= \bigtimes \cdot \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{\overline{\sum}}{2} \right)^k \\ &= \bigtimes \cdot \exp\left(\frac{1}{2} \overline{\sum}\right) \end{aligned} \quad (3.45)$$

Similarly, for $\underline{\sum}$, $\text{Wind} = -1/2$, thus

$$I\left(\underline{\sum}\right) = \bigtimes \cdot \exp\left(-\frac{1}{2} \underline{\sum}\right) \quad (3.46)$$

Example 3.2.3. Let's compute the first 2 terms of order 2 of the Kontsevich integral of a “hump” (the embedding of the unknot with four critical points)

$$H := \text{hump} \quad (3.47)$$

Degree 1 term is just $\bigcirc$. For degree 2 term, it suffices to compute the coefficient of the diagram $\bigcirc \otimes$. Clearly the only contribution to this term comes from configurations with chords between two critical points. Let's label the intersection points of t_1, t_2 -planes with H by $a_j, b_j, j = 1, 2, 3, 4$ as follows

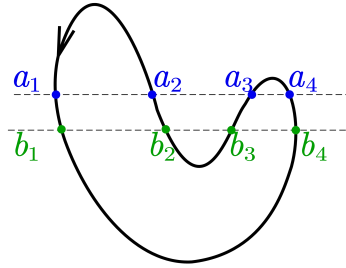


Figure 3.4: Points a_j, b_j

It's easy to see from fig. 3.5 the only contribution is from configurations with chords

$$\{a_{ij} := a_i a_j, b_{kl} := b_k b_l\}$$

where $i \in \{1, 2\}, j \in \{3, 4\}, k \in \{1, 4\}, l \in \{2, 3\}$.

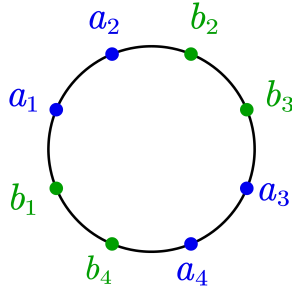


Figure 3.5: Corresponding chord diagram

Now the coefficient of $\bigcirc \otimes$ is

$$\begin{aligned} I &= \left(\frac{1}{2\pi\sqrt{-1}} \right)^2 \sum_{i,j,k,l} \int_0^1 (-1)^{i+j+k+l} \log' a_{ij} dt_1 \int_0^{t_1} \log' b_{kl} dt_2 \\ &= -\frac{1}{4\pi^2} \int d \log \left(\frac{a_{14} a_{23}}{a_{13} a_{24}} \right) d \log \left(\frac{b_{12} b_{34}}{b_{13} b_{24}} \right) \end{aligned} \quad (3.48)$$

Let

$$u = \frac{a_{13}a_{24}}{a_{14}a_{23}}, v = 1 - \frac{b_{12}b_{34}}{b_{13}b_{24}} \quad (3.49)$$

then u ranges from 0 to 1 and v ranges from 0 to

$$1 - \frac{a_{12}a_{34}}{a_{13}a_{24}} = \frac{a_{14}a_{23}}{a_{13}a_{24}} = u \quad (3.50)$$

thus, by example 3.1.1 at the beginning of this chapter,

$$I = \frac{1}{4\pi^2} \int_0^1 \frac{du}{u} \frac{dv}{1-v} = \frac{\zeta(2)}{4\pi^2} = \frac{1}{24} \quad (3.51)$$

As we have seen from example 3.2.3, I is a fake invariant since $I(H) \neq I(\bigcirc)$, but we will refine it into a genuine knot invariant $Z : \mathcal{K} \rightarrow \mathcal{A}$ by introducing a renormalization in section 3.3.

3.2.2 Basic Properties: Convergence, Functoriality and Monoidality

Proposition 3.2.4. $I(T)$ converges absolutely for any strict Morse tangle T .

Proof. The only problem of convergence is that tangles allow critical points, which means it might happen that $z_j - z'_j \rightarrow 0$ near some critical value. If for a configuration in P and a critical point c with value t_c , we have $t_j < \dots < t_i < t_c < t_{i-1}$, then either one of the points z_i, z'_i is not on the arc attaching the critical point c , or z_i, z'_i contribute to an isolated chord in D_P .

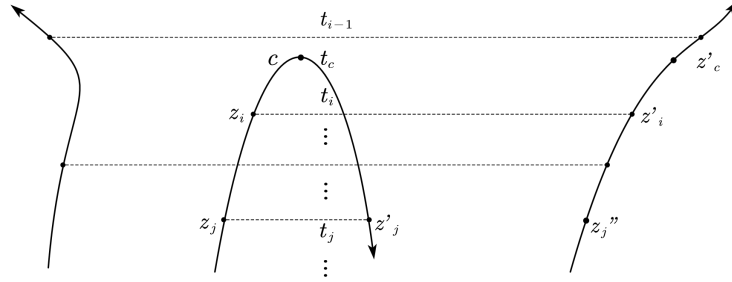


Figure 3.6: Convergence of the integral

Now since $z_i - z'_i$ is bounded below, the integral

$$\int_{t_i < t_c} |d \log(z_i - z'_i)| \leq C \int_{t_i < t_c} |d(z_i - z'_i)| < C_i |z_{i+1} - z'_{i+1}| \quad (3.52)$$

Where C_i is a uniform constant that only depends on the embedding of T near t_c . Inductively, assume

$$\int_{t_j < t_{j-1} < \dots < t_i < t_c} \bigwedge_{k=j-1}^i |d \log(z_k - z'_k)| \leq C_{j-1} |z_j - z'_j| \quad (3.53)$$

This would guarantee the integral

$$\begin{aligned} \int_{t_{j+1} < t_j < t_{j-1} < \dots < t_i < t_c} \bigwedge_{k=j}^i |\mathrm{d} \log(z_k - z'_k)| &\leq \int_{t_{j+1} < t_j} C_{j-1} |\mathrm{d} z_j - \mathrm{d} z'_j| \\ &\leq C_j |z_{j+1} - z'_{j+1}| \end{aligned} \quad (3.54)$$

Inductively, the integral converges absolutely as an improper integral on Δ^n by Tonelli's theorem, and we can integrate it iteratively in any order. $\square$

The first property of I is the multiplicativity. Here we only prove the result for the special embedding of product tangle, since we haven't shown the isotopy invariance.

Proposition 3.2.5. *The Kontsevich integral for tangles is multiplicative*

$$I(T_1 \cdot T_2) = I(T_1) \cdot I(T_2) \quad (3.55)$$

whenever $T_1 \cdot T_2$ is defined.

Proof. Note that

$$\begin{aligned} &\sum_P (-1)^{\downarrow P} D_P \bigwedge_{k=1}^n \omega_k(t_k) \\ &= \sum_{P_1 \cdot P_2 = P} \left((-1)^{\downarrow P_1} D_{P_1} \bigwedge_{i=1}^m \omega_i(t_i) \right) \cdot \left((-1)^{\downarrow P_2} D_{P_2} \bigwedge_{j=m+1}^n \omega_j(t_j) \right) \end{aligned} \quad (3.56)$$

where $\omega_k(t_k) = \mathrm{d} \log(z_k(t_k) - z'_k(t_k))$.

By Fubini's theorem and note that reparametrization does not change the integral, we have

$$\begin{aligned} I(T_1 \cdot T_2) &= \sum_n \sum_m \int_{0 < t_n < \dots < t_{m+1} < 1/2 < t_m < \dots < t_1 < 1} \sum_P (-1)^{\downarrow P} D_P \bigwedge_{k=1}^n \omega_k(t_k) \\ &= \sum_n \sum_m \left(\int_{1/2 < t_m < \dots < t_1 < 1} \sum_{P_1} (-1)^{\downarrow P_1} D_{P_1} \bigwedge_{i=1}^m \omega_i(t_i) \right) \\ &\quad \cdot \left(\int_{0 < t_n < \dots < t_{m+1} < 1/2} \sum_{P_2} (-1)^{\downarrow P_2} D_{P_2} \bigwedge_{j=m+1}^n \omega_j(t_j) \right) \\ &= I(T_1) \cdot I(T_2) \end{aligned} \quad (3.57)$$

$\square$

In particular, for knots, we have:

Corollary 3.2.6. *Let K_1, K_2 be two knots and $K_1 \# K_2$ be the connected sum of K_1, K_2 by cut and paste the bottom of K_1 and top of K_2 , then*

$$I(K_1 \# K_2) = I(K_1) I(K_2) \quad (3.58)$$

here the multiplication on the right hand side is multiplication (connected sum) in $\mathcal{A}$

Proof. Let $K'_1(K'_2)$ be the part of the knots after removing a neighborhood $\mathbb{U}(\mathbb{A})$ of the bottom of K_1 (top of K_2). Clearly $Z(\mathbb{U}) = \cup$ and $Z(\mathbb{A}) = \cap$, thus

$$I(K_1 \cdot K_2) = I(K'_1 \cdot \mathbb{U} \cdot \mathbb{A} \cdot K'_2) = I(K'_1) \cdot \cup \cdot \cap \cdot I(K'_2) \quad (3.59)$$

Thus

$$I(K_1 \# K_2) = I(K_1)I(K_2) \quad (3.60)$$

□

One might expect I also preserves the tensor product, but this is not true, since there will be contributions of chords connecting two diagrams. Nevertheless, we can diminish this bad behavior by introducing the ϵ -tensor: On a diagram of $T_1 \otimes T_2$ (Let's assume T_1 lies in $(-\infty, 0] + \sqrt{-1}\mathbb{R}$, and T_2 lies in $[1, \infty) + \sqrt{-1}\mathbb{R}$), we squeeze and compress the diagram using homothety with a ratio of $\epsilon \ll 1$ centered at 0 for T_1 , and centered at 1 for T_2 , and the result would be denoted by $T_1 \otimes_\epsilon T_2$. It turns out we have the following :

Proposition 3.2.7. *I satisfies the asymptotically monodality*

$$I(T_1 \otimes_\epsilon T_2) = I(T_1) \otimes I(T_2) + O(\epsilon) \quad (3.61)$$

Proof. Note that homothety does not change $Z(T_1)$ or $Z(T_2)$. The unwanted part in $Z(T_1 \otimes_\epsilon T_2)$ comes from chords that connect T_1 and T_2 . But since $|z_j - z'_j| \geq 1$, $\int |dz_j - dz'_j| \leq C\epsilon$ where C only depends on the initial embedding of T_1, T_2 . □

3.3 Isotopy Invariance

3.3.1 Isotopy Invariance within the Class of Morse Tangles

Theorem 3.3.1. *The Kontsevich integral I is invariant under deformations within the class of Morse tangles.*

Proof. For braids, the isotopy invariance follows directly from the homotopy invariance of the parallel transport. For tangles, we investigate the invariance step by step, just like in section 1.3.2.

1. Invariance under reparametrization. This is clear by the definition of I .
2. Invariance under horizontal deformations. A horizontal deformation is an isotopy that preserves all t -sections, and leaves a neighborhood of the t_c -plane fixed for any critical value t_c . Suppose $0 = t_{c_1} < \dots < t_{c_l} = 1$ are critical values and extremas of the tangle, we choose an $\epsilon \ll 1$, and consider $I_j = [t_{c_{j+1}} + \epsilon, t_{c_j} - \epsilon]$ and $J_j = [t_{c_j} - \epsilon, t_{c_j} + \epsilon]$, let $T_{[a,b]}$ be the restriction of the tangle T in $\mathbb{C} \times [a, b]$, then the multiplicativity proposition 3.2.5 implies that

$$Z(T) = \prod_j Z(T_{I_j}) Z(T_{J_j}) \quad (3.62)$$

But $Z(T_{I_j})$ is exactly the monodromy eq. (3.41) of some connection $\omega_{m,n}$ in eq. (3.37), horizontal deformations in I_j do not change it since the connection is flat.

3. Invariance under movements of critical points. A movement of a critical point is isotopy that is identity outside a neighborhood of the critical point, and does not introduce new critical points. We can first use horizontal deformations to squeeze the arc near the critical point into a “sharp needle” that lies in a tubular neighborhood whose horizontal cross-sections have diameters of less than ϵ (fig. 3.7) and intersect the needle twice.

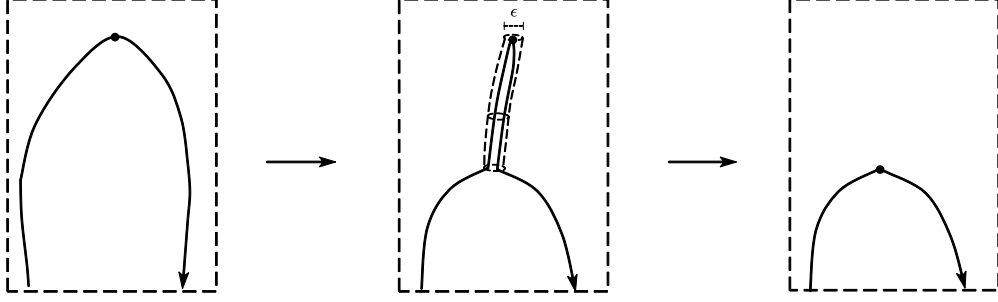


Figure 3.7: Movement of a critical point

Let's inductively show that the difference of integral on this diagram with the diagram that deletes this needle is $O(\epsilon)$ as $\epsilon \rightarrow 0$, just like the proof of convergence. Consider configurations so that $t_k < \dots < t_i < t_c < t_{i+1}$, then we only need to consider chords that one of the endpoints z_i, z'_i , say z_i , is not on the needle, otherwise it contributes to an isolated chord. But we can pair the chords z_i, z'_i with another configuration which is the same elsewhere except with chords z_i, z''_i , where z''_i is the other intersection point of the t_i -plane with the needle. The integration of these two terms has opposite signs, and their difference is clearly $O(\epsilon)$

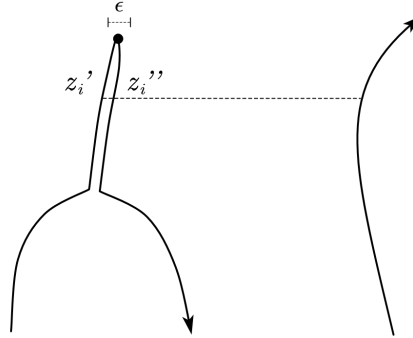


Figure 3.8: Contribution of the needle

Now, we can continue this process if next chords $t_{i+1}, \dots, t_k$ also connect the needle with other parts of the tangle. If not, then we can bound their contribution by the same inductive method as in proposition 3.2.4. Again, this is $O(\epsilon)$. Since ϵ is arbitrary, I remains unchanged if we remove this needle.

These three steps show that I is an invariant of Morse tangles. □

Remark 3.3.2. Kontsevich’s original proof [19] of invariance under horizontal deformation is using Stokes theorem for the iterated integral on the simplex $\{t_a < t_n < \cdots < t_1 < t_b\}$. See also [21] for details. There is also an elementary proof in [14], which essentially verifies the fact that formal KZ connection eq. (3.35) is flat, albeit in a disguised manner.

3.3.2 Changing the Number of Critical Points

Now let’s focus on knots. Although I is not invariant under the move that deletes or insertss critical points, we can predict the contribution of the added pair of critical points. Let H be the hump defined in example 3.2.3. Use the preceding deformations to slide the “humpy part” along the knot to the top of the knot and denote the resulting knot K_H , then $K_H = H \cdot K$.

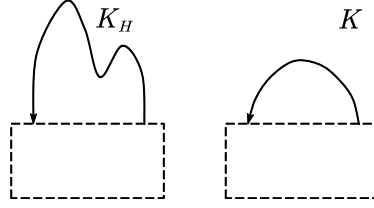


Figure 3.9: $K_H = H \cdot K$

since I preserves connected sum by corollary 3.2.6, it follows that:

$$I \left(\text{Wavy line} \right) = I(H) \cdot I \left(\text{Straight line} \right) \quad (3.63)$$

Any isotopy of a tangle can be approximated by a sequence of isotopies within the class of Morse tangles(the first 3 moves) and deletions or insertions of critical points(the last move). Note that $I(H)$ is invertible in the algebra $\mathcal{A}$ since the degree 1 term is $\bigcirc$, with the inverse given by

$$\bigcirc + \sum_{j=1}^{\infty} (-1)^j (I(H) - \bigcirc)^n \quad (3.64)$$

we conclude that

Theorem 3.3.3 (Kontsevich[19]). *The integral*

$$Z(T) = \frac{I(T)}{(I(H))^{c/2-1}} \quad (3.65)$$

is invariant under arbitrary ambient isotopy of the tangle, where c is the number of critical points.

The choice of coefficients of $I(H)$ in the denominator is made so that

$$I(\bigcirc) = \bigcirc. \quad (3.66)$$

and Z is multiplicative:

$$Z(K_1 \# K_2) = Z(K_1)Z(K_2) \quad (3.67)$$

for any knots K_1, K_2 .

3.4 Universality, Reality and Group-like Property of the Kontsevich Integral

In this section, we end our proof of the fundamental theorem 2.3.1, which boils down to examining that

Proposition 3.4.1. *If K is an n -singular knot, $Z(\partial^n K)$ only has order $\geq n$, and the n -th term is exactly the chord diagram D_K associated with K .*

Proof. Since $I(H) = \bigcirc + (\text{higher order terms})$ and $\bigcirc$ is unit in $\mathcal{A}$, it suffices to prove this for I .

Let J be a labeling set of double points and suppose the resolutions of K are identical except in neighborhoods $\{U_j, j \in J\}$, assume further that U_{j+1} is lower than U_j . Resolve K at $j \in J$: $\partial^n K = \partial^{n-1} K_j^+ - \partial^{n-1} K_j^-$.

Now, if a configuration P has no chord in a neighborhood U_j , then we can pair a term of $\partial^{n-1} K_j^+$ with a term of $\partial^{n-1} K_j^-$ so that they are identical except in U_1 , thus their contribution to the integral cancel out. This implies that if the number of chords is less than n , then the integral is zero. When the number of chords is n , the only contribution comes from configurations whose n chords lie in the n distinct neighborhoods $\{U_j\}$. Their chord diagram is D_K . Each iteration of the integral is nothing but the winding number $\text{Wind}(j, \pm)$ of the strands in U_j with $+$ or $-$ crossings. For example, if K_f is a knot in the complete resolution of K with crossings determined by $f : J \mapsto \{\pm\}$, then

$$Z(K_f) = \prod_{i=1}^n \text{Wind}(i, f_i) D_K. \quad (3.68)$$

But $\text{Wind}(i, +) - \text{Wind}(i, -) = 1$, thus

$$\begin{aligned} Z(K) &= \sum_f (-1)^{\#f^{-1}(-)} Z(K_f) \\ &= \sum_f (-1)^{\#f^{-1}(-)} \prod_{i=1}^n \text{Wind}(i, f_i) \\ &= \prod_i (\text{Wind}(i, +) - \text{Wind}(i, -)) D_K \\ &= D_K. \end{aligned} \quad (3.69)$$

This completes the proof of the fundamental theorem. $\square$

Although so far we have been working over the base field $\mathbb{C}$ in this chapter, the fundamental theorem is valid for any base field $\mathbb{F}$ of characteristic zero thanks to the rationality of Kontsevich integral:

Theorem 3.4.2 (Le-Murakami[20]). *For any knot K , the coefficients of the Kontsevich integral $Z(K)$ in a basis of $\mathcal{A}$ consisting of chord diagrams are rational.*

The proof of this theorem relies on the existence of a rational associator, and is highly non-trivial and rather involved, so we shall not discuss it here. However, we can easily show that coefficients of Z are real.

Theorem 3.4.3. *All coefficients of Z in a basis of chord diagrams are real.*

Proof. Again it suffices to prove this for I . Consider the knot $\bar{K}$ obtained from K by $t \mapsto 1 - t, z \mapsto \bar{z}$ namely the reflections about the $t = 1/2$ plane and the $\mathbb{R} \times \mathbb{R}_t$ plane, then $\bar{K}$ is actually obtained from K by rotation of 180 degrees around the $\{t = 1/2, z \in \mathbb{R}\}$ axis. Now every string in a configuration P with n pairs reverses the orientation and contributes to the sign $(-1)^{2n-\downarrow_P} = (-1)^{\downarrow_P}$. Note that $d \log(\bar{z}_j - \bar{z}'_j) = \overline{d \log(z_j - z'_j)}$, we have

$$\begin{aligned} \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int \bigwedge_{j=n}^1 d \log(\bar{z}_j - \bar{z}'_j) &= \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int_0^1 (-1)^{\frac{n(n+1)}{2}} \bigwedge_{j=n}^1 \overline{d \log(z_j - z'_j)} \\ &= \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int_0^1 (-1)^{\frac{n(n+1)}{2}} \cdot (-1)^{\frac{n(n-1)}{2}} \bigwedge_{j=1}^n \overline{d \log(z_j - z'_j)} \\ &= \overline{\left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int_0^1 \bigwedge_{j=1}^n d \log(z_j - z'_j)}, \end{aligned} \quad (3.70)$$

where the sign $(-1)^{n(n+1)/2}$ comes from change of variables $t_j \mapsto 1 - t_{n+1-j}$ between the n -simplices. It follows that $Z(K) = Z(\bar{K}) = \overline{Z(K)}$, and coefficients of $Z(K)$ are real. $\square$

Another interesting property of $Z : \mathcal{K} \rightarrow \hat{\mathcal{A}}$ is that it's in fact a bialgebra morphism. It's an algebra morphism because it preserves the unit by eq. (3.66) and connected sum by eq. (3.67). It clearly preserves counit because the degree 0 term of $Z(K)$ for any knot K is always $\bigcirc$. The only property left to verify is:

Proposition 3.4.4. *The Kontsevich integral of a strict Morse tangle T is group-like:*

$$\delta I(T) = I(T) \otimes I(T) \quad (3.71)$$

Proof. Let's focus on a configuration P with $|P|$ pairs, $\delta(D_P) = \sum_{J \subset P} D_J \otimes D_{P \setminus J}$ by eq. (2.30) where the sum is taken over all subconfigurations J of P . Now the coefficient c_J of the contribution of configuration J to $I(K)$ is an integral over $\Delta_J = \{0 \leq t_{|J|} \leq \dots \leq t_1 \leq 1\}$, and $c_{P \setminus J}$ is an integral over $\Delta_{P \setminus J}$. $c_J \cdot c_{P \setminus J}$ is an integral over $\Delta_J \times \Delta_{P \setminus J}$, which can be splitted into a sum of integration over $\binom{|P|}{|J|}$ $|P|$ -simplices, so

$$c_P = \binom{|P|}{|J|} c_J \cdot c_{P \setminus J} = \sum_{J' \cup J'' = P} c_{J'} c_{J''}. \quad (3.72)$$

As a consequence:

$$\begin{aligned} \delta I(T) &= \sum_P c_P \delta(D_P) \\ &= \sum_P \sum_{J \subset P} c_P D_J \otimes D_{P \setminus J} \\ &= \sum_{J', J''} c_{J'} c_{J''} D_{J'} \otimes D_{J''} \\ &= I(T) \otimes I(T). \end{aligned} \quad (3.73)$$

$\square$

In particular, for any knot K , $Z(K) = I(K)/(I(H))^{c/2-1}$ is also group-like in $\hat{\mathcal{A}}$. As a corollary,

Corollary 3.4.5. *Both Z and Z^* are bialgebra morphisms. We have $\widehat{\mathcal{K}} \cong \hat{\mathcal{A}}$ and $\mathcal{V} \cong \mathcal{W}$ as bialgebras.*

Chapter 4

Configuration Space Integrals

In this chapter, we introduce the other universal finite type invariants: the Chern-Simons series constructed as a power series of the configuration space integral. Configuration space integrals were motivated by Chern-Simons theory in Guadagnini, Martellini, and Mintchev [16], Bar-Natan [6] and Kontsevich [19]. In fact, these integrals appear earlier before the Kontsevich integral, but the analytic difficulties are harder to resolve. Based on compactification techniques developed by Fulton and MacPherson in [15] as well as Axelrod and Singer [3], Bott and Taubes studied the degree 2 integrals from a rigorous topological viewpoint [8]. Altschuler and Freidel [1] and D. Thurston in his thesis [28] independently generalized the result of Bott and Taubes to all degrees. It turns out the perturbative Chern-Simons series is actually another universal finite type invariant of knots.

4.1 Bott-Taubes Integration and Chern-Simons Series

The Euclidean configuration space introduced in section 1.2 is not enough for our purpose, we need a notion of restricted configuration space where some points constrained on a knot $K : \mathbb{S}^1 \rightarrow \mathbb{R}^3$ and other points allowed to move freely in $\mathbb{R}^3$. Let V be labeling sets and $U \subset V$, and fix an isotopy class $[i]$ of maps $U \hookrightarrow K(\mathbb{S}^1)$ (so that we can restrict our attention to a connected component), define

$$\text{Conf}_{V,U}(\mathbb{R}^3, K) = \left\{ (x_v)_{v \in V} \in \text{Conf}_V(\mathbb{R}^3) : (U \rightarrow K(\mathbb{S}^1), u \mapsto x_u) \in [i] \right\} \quad (4.1)$$

Each Jacobi diagram Γ gives rise to a configuration space like this, simply let V be the set of vertices of Γ , U be the set of univalent vertices and $[i]$ be the isotopy class of them, the resulted space is denoted by $C(K, \Gamma)$. Note that here I didn't use the orientation of trivalent vertices of Γ .

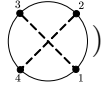
Let's try (harder) again to generalize the linking number to knots, which we have failed to do in section 1.3. This time we consider 4 points, labeled by 1, 2, 3, 4 on $K(\mathbb{S}^1)$, again there are maps

$$\phi_{ij} : \text{Conf}_4(\mathbb{S}^1) \rightarrow \mathbb{S}^2, (\theta_i, \theta_j) \mapsto \frac{K(\theta_j) - K(\theta_i)}{|K(\theta_j) - K(\theta_i)|} \quad (4.2)$$

But since we have failed to produce invariants when we choose maps that i is incident to j , we consider

$$\phi_{13} \times \phi_{24} : \text{Conf}_4(\mathbb{S}^1) \rightarrow \mathbb{S}^2 \times \mathbb{S}^2 \quad (4.3)$$

this time. We use chords to connect i and j , then this map factors through

$$\text{Conf}(K) : \text{Conf}_4(\mathbb{S}^1) \rightarrow \text{Conf}_4(\mathbb{R}^3, K) = C(K, \text{diagram}) \quad (4.4)$$


and we have

$$\Phi_{13} \times \Phi_{24} : C(K, \text{diagram}) \rightarrow \mathbb{S}^2 \times \mathbb{S}^2 \quad (4.5)$$


This is a map from a 4 dimensional manifold to a 4 dimensional manifold, so it's attempting to write

$$I(K, \text{diagram}) = \int_{C(K, \text{diagram})} \Phi^* \omega \quad (4.6)$$


for an orientation form ω on $\mathbb{S}^2 \times \mathbb{S}^2$ so that $\int_{\mathbb{S}^2 \times \mathbb{S}^2} \omega = 1$, and we say this is the integral

associated to the diagram .

This is illustrating, since it's now natural to define similar maps to any Jacobi diagrams.

In degree 2, there is another Jacobi diagram: . The map

$$\Phi_{14} \times \Phi_{24} \times \Phi_{34} : C(K, \text{diagram}) \rightarrow \mathbb{S}^2 \times \mathbb{S}^2 \times \mathbb{S}^2 \quad (4.7)$$


is a map between 6 dimensional manifolds.

In general, for a Jacobi diagram with edges labeled by E , univalent vertices labeled by U and trivalent vertices labeled by T , the map

$$\Phi : C(K, \Gamma) \rightarrow \mathbb{S}^E \quad (4.8)$$

that sends a point in $C(K, \Gamma)$ to the direction vectors of its edges is a map between manifolds of dimension $\#U + 3\#T = 2\#E$. We define

$$I(K, \Gamma) = \int_{C(K, \Gamma)} \Phi^* \omega \quad (4.9)$$

where ω is an orientation form on $(\mathbb{S}^2)^E$ with $\int_{(\mathbb{S}^2)^E} \omega = 1$.

There is still one subtlety: we need to choose a canonical orientation of the manifolds so that the sign of $I(K, \Gamma)$ is well defined. The orientations of the trivalent vertices (which is part of the data of the diagram by definition!) determine an order of three half edges at a trivalent vertex up to even permutations, and this provides a preferred correspondence between the order of $U \sqcup 3T$ and the order of half edges up to even permutations. Giving these edges arbitrary orientations and an order of E equivalently determines an order of half edges, which in turn gives an orientation of $M^U \times (\mathbb{R}^3)^T$. Now $C(K, \Gamma)$ as an open subset of $M^U \times (\mathbb{R}^3)^T$ is thus naturally oriented.

We have to show this is independent of our choices of the order of E and orientations of edges. To do this, we first require the form ω to be antisymmetric under the antipodal maps on each $\mathbb{S}^2$ and invariant under any permutation of E . This can be achieved by setting $\omega = \bigwedge_{e \in E} \omega_e$ where each ω_e is antisymmetric under the antipodal map. For example, we can set each ω_e to be the canonical $SO(3)$ invariant form

$$\omega_e = \frac{1}{4\pi} (z dx \wedge dy + x dy \wedge dz + y dz \wedge dx) \quad (4.10)$$

(expressed in standard coordinates of $\mathbb{R}^3$) and their wedge product $\omega = \bigwedge^E \omega_e$ on $(\mathbb{S}^2)^E$.

Changing the orientation of an edge swaps the order of two vertices but also results in a sign change of $\Phi^*(\omega)$ as our ω_e is antisymmetric under antipodes, changing the order of two edges doesn't change $\Phi^*\omega$ since we have chosen ω to be invariant under permutations of E . As a result, the definition of $I(K, \Gamma)$ is independent of the choice of the orientations and the order of edges. It's important to note that although $I(K, \Gamma)$ depends on the orientation of trivalent vertices, $I(K, \Gamma)\Gamma$ as an element in the algebra $\mathcal{A}$ modulo $1T, STU$ relations doesn't.

As we shall see later, which form ω we use to integrate doesn't really matter, so the sign of $I(K, \Gamma)$ is now well defined.

At degree 1, the integral $I(K, \textcirclearrowleft)$ is just the Gauss integral we have tried to generalize in section 1.3. This is not a knot invariant.

The first nontrivial integral invariant appears at degree 2:

$$I = \frac{1}{4} I(K, \textcirclearrowleft) + \frac{1}{3} I(K, \textcirclearrowright) \quad (4.11)$$

Bott and Taubes[8] were the first to study it from a topological perspective.

Note that the coefficients 4 and 3 in denominators are numbers of automorphisms of the corresponding graphs. eq. (4.11) can be generalized to all orders. The perturbative Chern-Simons series of a knot K is defined to be

$$Z_{CS}(K) = \sum_{\Gamma} \frac{I(K, \Gamma)\Gamma}{\#\text{Aut}\Gamma} \in \hat{\mathcal{A}} \quad (4.12)$$

where the sum is taken over all Jacobi diagrams forgetting the orientations (It doesn't matter which representative of Γ we are using since $I(K, \Gamma)\Gamma$ is independent of the choice of vertex orientation). To avoid using $\#\text{Aut}\Gamma$, we can consider the sum over the labeled graph, since there are exactly $(\#V)!/\#\text{Aut}\Gamma$ ways to label Γ :

$$Z_{CS}(K) = \sum_{\text{labeled } \Gamma} \frac{I(K, \Gamma)}{(\#V)!} \quad (4.13)$$

Theorem 4.1.1 (D. Thurston[28], Altschuler-Freidel[1]). *$Z(K) := Z_{CS}(K)$ defined for representative K with zero Gauss integral $I(K, \textcirclearrowleft) = 0$ is a universal finite type knot invariant.*

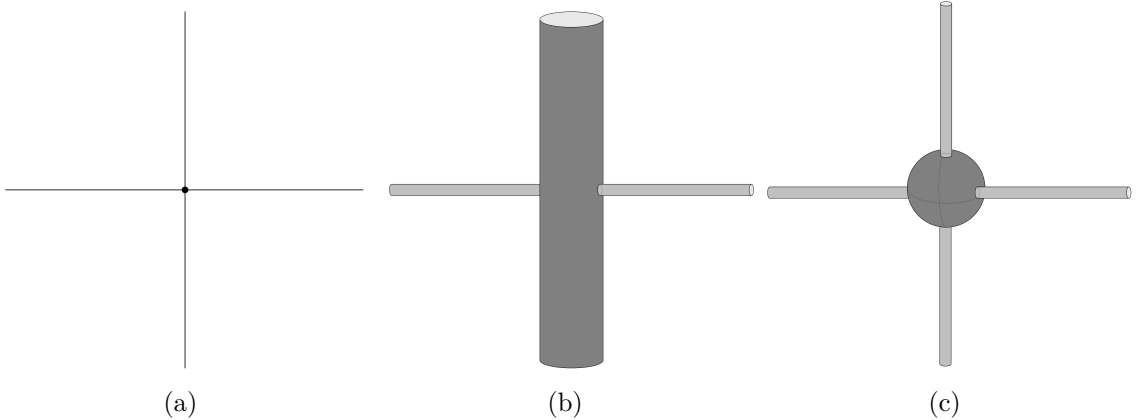
In order to show the integral $I(K, \Gamma)$ over $C(K, \Gamma)$ do converges, as in the case of linking number in section 1.3, we need to compactify it into some space $C[K, \Gamma]$, so that the edge function Φ can be smoothly extended. This will be done in the next section. This done, we can use Stokes theorem to show that these integrals are invariant under isotopy of knot.

4.2 Compactification of Configuration Space

There is a canonical compactification of configuration spaces known as the Fulton-MacPherson compactification in algebraic geometry, and was first introduced by Axelrod and Singer in the case of differential geometry[3]. It is functorial with respect to the immersions of smooth manifolds. Here we briefly review another construction of compactification of Dylan Thurston presented in [28].

Let's first look at $\text{Conf}_V(M) \subset (M)^V$ where V is a labeling set so that $\#V = n$. We are going to attach boundaries to $\text{Conf}_V(M)$, which records the infinitesimal moves as some points collide. This is done by blowing up along those diagonals $L \subset M' := M^V$, namely to remove L and replace it with the unit normal bundle SNL . The result is a manifold with corner $Bl(M', L)$, which is a union of M' and lower dimensional manifolds called the strata of M' . The canonical projection map $Bl(M', L) \rightarrow M'$ allows us to define blowup by some universal property([24]).

To blow up along a diagonal Δ_A where all x_a coincide for $a \in A \subset V$, we replace a tubular neighborhood $\Delta_A \times B_\epsilon$ with $\Delta_A \times \bar{B}_{\epsilon/2}$, and we construct the canonical projection by sending $\Delta_A \times B_{\epsilon/2}$ to $\Delta_A \times B_\epsilon$ and $\Delta_A \times \partial B_{\epsilon/2}$ to Δ_A . The order of the blowup is significant, it's important that we blow up the sub-diagonal Δ_A before Δ_B whenever $B \subset A$. The compactification $\text{Conf}_V[M]$ is constructed by successively blowing up the diagonals of M^n , starting from the main diagonal $\Delta = \{(x, \dots, x), x \in M\}$, and perform blowups along Δ_C for all $\#C = n - 1$, and then for $\#C = n - 2, \dots$. When we blow up Δ_B , we mean blowing up the closure of Δ_B minus all of its subdiagonals(which have already been blown up). Proceeding this way, the result of the blowup is independent of the order at each stage for fixed $\#C$. For example, if we blow up two intersecting lines L_1, L_2 in $\mathbb{R}^3$, then different orders would produce different results. But if we have already blown up their intersection point, then the order that we blow up L_1, L_2 doesn't matter. For example, blowing up the lines $L_1, L_2 \in \mathbb{R}^3$ in (a) will depend on the order, but if we have blown up the point $L_1 \cap L_2$, then the order doesn't matter as in (c).



In fact, whether or not successive blowups along submanifolds $L_i, i = 1, \dots, r$ depend on the order performed is related to the way these submanifolds intersect in M . If their intersection is empty, then clearly the order doesn't matter. If they do intersect, but after a small arbitrary move they still intersect nontrivially, then we say that they intersect generically.

This is equivalent to say

$$\bigoplus_{i=1}^r \mathcal{T}_x L_i \rightarrow (\mathcal{T}_x M)^r / \Delta \quad (4.14)$$

is surjective where the diagonal $\Delta = \{(v, \dots, v) \in (\mathcal{T}_x M)^r, v \in \mathcal{T}_x M\}$. If $r = 2$, then intersecting generically is equivalent to intersecting transversely, but in general generical intersection is nicer than pairwise transverse intersection, although in fact the latter also implies the independence of the order.

If $L_1, \dots, L_n$ intersect generically at x , we can choose a chart so that they are hyperplanes in a small neighborhood U of $x \in M$. By the linear algebra lemma proposition B.0.1 in the appendix, we have a decomposition $U = U' \times U_1 \times \dots \times U_n, x = x' \times x_1 \times \dots \times x_n$ so that the blowup of $L_k = U' \times U_1 \times \dots \times x_k \times \dots \times U_k$ in M is $U' \times U_1 \times \dots \times Bl(U_k, x_k) \times \dots \times U_n$. Thus, the result of successive blowups is $U' \times Bl(U_1, x_1) \times \dots \times Bl(U_n, x_n)$ and is independent of the order.

Now, in our process, if $C \cap D \neq \emptyset$, then $\Delta_C \cap \Delta_D = \Delta_{C \cup D}$ in M^n has been blown up, and the closure of Δ_C, Δ_D in this stage do not intersect. If $C \cap D = \emptyset$, then it's easy to see that in a local chart

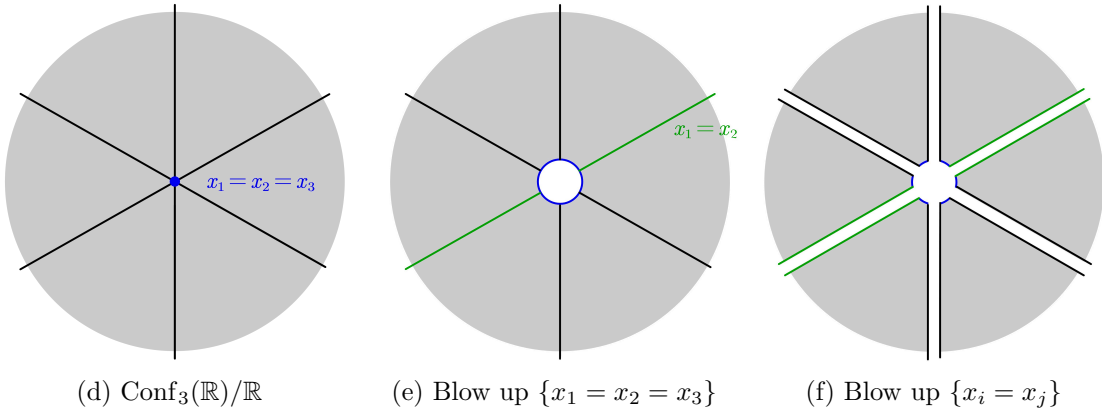
$$(a + \epsilon_1, \dots, a + \epsilon_{|C|}, \dots) = (\dots, b + \delta_1, \dots, b + \delta_{|D|}, \dots) \quad (4.15)$$

do have a solution for any $\epsilon_i, \delta_j \ll 1$. Thus, Δ_C and Δ_D intersect generically and the order of blowups doesn't matter.

For an embedding $N \rightarrow M$, the subdiagonals of N^V are precisely restrictions of diagonals of M^V , so the blowup is functorial.

To get a feel for what $\text{Conf}_V[M]$ looks like, we take a look at some examples.

Example 4.2.1. For $\text{Conf}_3[\mathbb{R}]$, we can draw this compactification in coordinates. Note that $\text{Conf}_3(\mathbb{R}) = \mathbb{R} \times \text{Conf}_2(\mathbb{R} \setminus 0)$, the compactification modulo $\mathbb{R}$ can be visualized as:



Example 4.2.2. For $\text{Conf}_2(\mathbb{S}^1)$, the compactification is just adding two circles and $\text{Conf}_2[\mathbb{S}^1]$ is just an annulus $\mathbb{S}^1 \times [0, 1]$. For $\text{Conf}_n[\mathbb{S}^1]$, let's fix an isotopy class (a cyclic order) so that only look at one connected component:

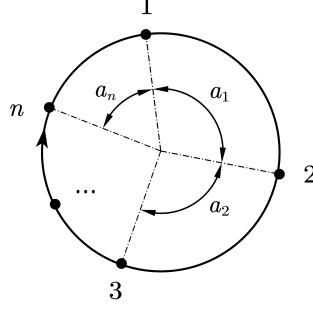


Figure 4.1: $\text{Conf}_n(\mathbb{S}^1) = \mathbb{S}^1 \times \Delta^{n-1}$

we reparametrize the points so $\text{Conf}_n(\mathbb{S}^1) = \mathbb{S}^1 \times \text{Int}\Delta_{n-1}$ where $\text{Int}\Delta_{n-1}$ is the interior of a $(n-1)$ -simplex

$$\text{Int}\Delta^{n-1} = \left\{ (a_1, \dots, a_n) : a_i > 0, \sum_{i=1}^n a_i = 1 \right\} \quad (4.16)$$

To compactify $\mathbb{S}^1 \times \Delta^2$, we first blow up the vertices of Δ^2 (which are actually the same point in $(\mathbb{S}^1)^3$), the result is attaching three pieces of a torus (since we restrict on one connected component) labeled by $(123), (231), (312)$ with (ijk) means points labeled by i, j, k collide, which graphically looks like “chopping off” the three corners and paste a boundary along the three faces. Then, blow up the diagonals labeled by $(12), (13), (23)$ by adding 3 faces to the rest of the sides. Thus $\text{Conf}_2[\mathbb{S}^1] = \mathbb{S}^1 \times W_3$ where W_3 is a hexagon. In general,

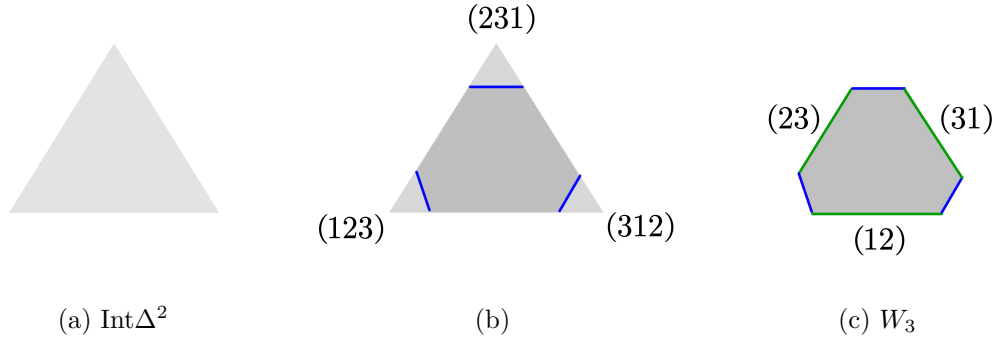


Figure 4.2: W_3

$\text{Conf}_n[\mathbb{S}^1] = \mathbb{S}^1 \times W_n$ where W_n is a polyhedra. W_4 looks like the following. The faces of W_n are labeled by subsets $\{S \subset (1, \dots, n), \#S \geq 2\}$. The ridges (lower-dimensional boundaries) are labeled by “non-associated words” for example $((12)3)(45)$. The codimension of the ridges equals the number of the parentheses. We mainly concern codimension 1 boundaries which will appear in Stokes formula. The faces that record the limit configuration of two points collapse are called principle faces, colored green in figs. 4.2 and 4.3. They are produced at the final stage of compactification. The rest of the faces are called the hidden faces. The faces blown up at the first stage are called anomalous faces, they are colored blue in figs. 4.2 and 4.3

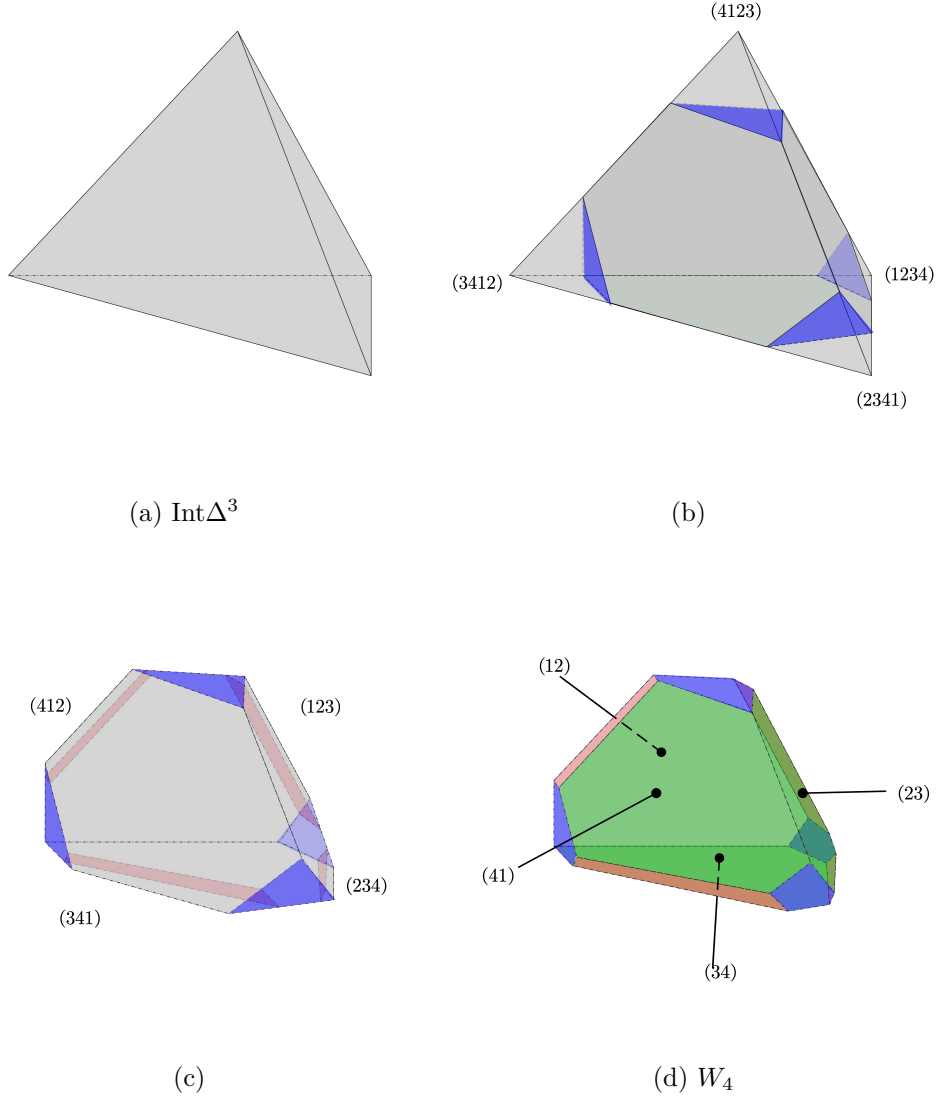


Figure 4.3: W_4

In general, the stratum of $\text{Conf}_V[\mathbb{R}^k]$ are labeled by nested sets $\mathfrak{S} \subset 2^A$ so that each $B, C \in \mathfrak{S}$ satisfy $\#B, \#C \geq 2$ and exactly one of: $B \cap C = \emptyset$, $B \subset C$ or $C \subset B$. The codimension of $\mathfrak{S}$ is $\#\mathfrak{S}$. In particular, the codimension 1 faces, which are most important for us since they contribute to the integration on the boundary, are labeled by subsets of A .

For us, we consider compactification of $\text{Conf}_{V,\infty}(\mathbb{S}^3) = \text{Conf}_V(\mathbb{R}^3)$, where we also blow up ∞ at each stage, and we simply write the compactification as $\text{Conf}_V[\mathbb{R}^3]$. The configuration space $\text{Conf}_{V,U}(\mathbb{R}^3)$ is embedded in $\text{Conf}_V[\mathbb{R}^3]$, and we define $\text{Conf}_{V,U}[\mathbb{R}^3]$ to be the closure of $\text{Conf}_{V,U}(\mathbb{R}^3)$. The maps $\Phi_{ij} : \text{Conf}_{V,U}(\mathbb{R}^3) \rightarrow \mathbb{S}^2$ extends to this space smoothly. This implies the convergence of the integral $I(K, \Gamma)$.

4.3 Integration Along the Fiber

Recall that for a vertically compactly supported k -form ω on an oriented fiber bundle $\pi : E \rightarrow M$ whose fibers are n -dimensional manifolds, we can integrate it along the fiber

$$(\pi_*\omega)_p(v_1, \dots, v_{k-n}) = \int_{\pi^{-1}(p)} \omega(\widetilde{v}_1, \dots, \widetilde{v}_{k-n}, -, \dots, -) \quad (4.17)$$

where $\widetilde{v}_i$ are lifts of v_i . One can easily deduce an explicit local formula by decomposing ω to two types of forms in local charts. By de Rham theorem and Fubini's theorem, it is also equivalent to the unique form $\pi_*\omega$ determined by

$$\int_c \pi_*\omega = \int_{\pi^{-1}(c)} \omega \quad (4.18)$$

for every smooth $(k-n)$ -chain $c \in M$.

If the fiber F is a manifold without boundary, π_* commutes with exterior differential d . But if the fiber is a manifold with corners, then

Proposition 4.3.1 (Fiberwise Stokes formula). *Let $\pi : E \rightarrow M$ be an oriented fiber bundle whose fiber F is n -dimensional smooth manifold with corners, let $\partial\pi : \partial E \rightarrow M$ and ω be a k -form on E with compact vertical support*

$$d\pi_*\omega = \pi_*d\omega + (-1)^{k-n}(\partial\pi)_*\omega \quad (4.19)$$

Proof. On any trivialised $E_c := \pi^{-1}c \cong c \times F$ where c is a smooth $(k-n+1)$ -chain in M , using Fubini theorem and Stokes formula for chains

$$\begin{aligned} \int_c \pi_*(d\omega) &= \int_{E_c} d\omega \\ &= \int_{\partial E_c} \omega \\ &= \int_{E_{\partial c}} \omega + \int_{(\partial E)_c} \omega \\ &= \int_{\partial c} \pi_*\omega + (-1)^{k-n+1} \int_c \partial\pi_*\omega \\ &= \int_c (d\pi_*\omega - (-1)^{k-n} \partial\pi_*\omega) \end{aligned} \quad (4.20)$$

The sign $(-1)^{k-n+1}$ comes from the fact that the orientation of $(\partial E)_x$ for $x \in c$ differs from the orientation inherited from $c \times \partial E \subset \partial(c \times E)$ by a factor $(-1)^{\dim c}$.

We also give a precise elementary proof on local charts. After choosing a partition of unity, it suffices to consider ω in a local coordinate $U \times H = \{(x, t)\}$ as linear combination of two types of forms: $\omega_\alpha = f(x, t)dx^I \wedge dt^J$ and $\omega_\beta = g(x, t)dx^{I'} \wedge dt^{J'}$, where $\#J = n$ while $\#J' < n$. Now by definition

$$\begin{aligned} d\pi^*\omega_\alpha &= d\left(\int_H f(x, t)dt^J\right) dx^I \\ &= \sum_i \left(\int_H \frac{\partial}{\partial x^i} f(x, t)dt^J\right) dx^i \wedge dx^I \end{aligned} \quad (4.21)$$

on the other hand

$$\pi_* d\omega_\alpha + (-1)^{k-n}(\partial\pi)_*\omega_\alpha = \sum_i \left(\int_H \frac{\partial}{\partial x^i} f(x, t) dt^J \right) dx^i \wedge dx^I \quad (4.22)$$

Meanwhile $\pi_*\omega_\beta = 0$, and if $\#J' = n - 1$ then

$$\begin{aligned} \pi_* d\omega_\beta &= \pi_* \left(\sum_j \frac{\partial}{\partial t^j} g(x, t) dt^j \wedge dx^{I'} \wedge dt^{J'} \right) \\ &= (-1)^{\#I'} \left(\int_H \sum_j \frac{\partial}{\partial t^j} g(x, t) dt^j \wedge dt^{J'} \right) dx^{I'} \\ &= (-1)^{\#I'} \left(\int_H d \left(g(x, t) dt^{J'} \right) \right) dx^{I'} \\ &= (-1)^{\#I'} \left(\int_{\partial H} g(x, t) dt^{J'} \right) dx^{I'} \\ &= -(-1)^{k-n}(\partial\pi)_*\omega_\beta \end{aligned} \quad (4.23)$$

If $\#J' < n - 1$, then everything vanishes. $\square$

Now, fix a Jacobi diagram Γ with edges labeled by E , our configuration spaces $C[-, \Gamma] = \bigcup_{K \in \mathfrak{K}} C[K, \Gamma] \subset \mathfrak{K} \times \text{Conf}_V[\mathbb{R}^3]$ can be viewed as a fiber bundle over the space of knots $\mathfrak{K}$ with fiber at $K \in \mathfrak{K}$ being $C[K, \Gamma]$.

$$\begin{array}{ccc} C[-, \Gamma] & \xrightarrow{\Phi} & (\mathbb{S}^2)^E \\ \pi \downarrow & & \\ \mathfrak{K} & & \end{array} \quad (4.24)$$

$I(K, \Gamma)$ is exactly $\pi_*(\Phi^*\omega)_K$, the integration of $\Phi^*\omega$ along the fiber $C(K, \Gamma)$ at K . We need to investigate if $I(-, \Gamma)$ is a locally constant function on $\mathfrak{K}$. Fiberwise Stokes formula proposition 4.3.1 allows us to investigate how $\pi_*\omega$ varies as a function on $\mathfrak{K}$. If we choose closed form ω , then the only thing to worry about is the variation of $(\partial\pi)_*\omega$, namely we need to investigate the integration on faces of $C[K, \Gamma]$.

Besides, it doesn't really matter which form in the cohomology class of $\Phi^*\omega$ we are using if the boundary integral always vanishes, since if μ is an $n - 1$ form

$$\begin{aligned} \pi_*(\Phi^*\omega + d\mu) - \pi_*(\Phi^*\omega) &= \pi_* d\mu \\ &= d\pi_*\mu - (-1)^{k-n}(\partial\pi)_*\mu \\ &= -(-1)^{k-n}(\partial\pi)_*\mu \end{aligned} \quad (4.25)$$

So one may feel free to choose whatever form one likes. A bump form on $(\mathbb{S}^2)^{\#E}$ will again give a combinatorial description of the integral, just like the case of the linking number, which leads to the rationality of the integral.

To avoid the consideration of the smooth structure on $\mathfrak{K}$ (Which can be done, see for example [10]), we consider a path $K^t, t \in [0, 1]$ in $\mathfrak{K}$, namely a specific isotopy $\mathbb{R}^3 \times [0, 1] \rightarrow$

$\mathbb{R}^3$. The restriction of configuration spaces on this path $\cup_t C(K^t, \Gamma)$ is an honest finite dimensional fiber bundle over $[0, 1]$:

$$\begin{array}{ccc} \cup_t C[K^t, \Gamma] & \xrightarrow{\Phi} & (\mathbb{S}^2)^{\#E} \\ \pi \downarrow & & \\ [0, 1] & & \end{array} \quad (4.26)$$

Then the integrals are

$$I(K^t, \Gamma) = \pi_*(\Phi^*\omega) \quad (4.27)$$

We want some integral

$$I = \sum_{\Gamma} c_{\Gamma} I(K^t, \Gamma) \quad (4.28)$$

stays constant as t varies where c_{Γ} is some coefficients. Stokes formula proposition 4.3.1 says if we can prove at every time

$$\sum_{\Gamma} c_{\Gamma} (\partial\pi)_*\omega = \sum_{\Gamma} c_{\Gamma} \int_{\partial C[K, \Gamma]} \Phi^*(\omega) \quad (4.29)$$

vanishes, then the integral I is indeed an invariant.

4.4 Invariance and Universality of the Integral

By virtue of the fiberwise Stokes theorem, it suffices to consider the integral on codimension 1 faces of the configuration space $C[K, \Gamma]$. As we have mentioned, they are indexed by subsets A ($\#A \geq 2$) of the set of vertices V . The faces where 2 vertices collide with each other are called principal faces. The remaining faces are hidden faces, among which we call a face where some connected components of trivalent vertices of the graph collapse an anomalous face (for primitive graphs, this corresponds to the case when all points come together).

Proposition 4.4.1. *Principal faces cancel each other.*

Proof. We sketch this proof case by case.

1. A face where two trivalent vertices v_1, v_2 connected by a dashed line collide is isomorphic to

$$\text{Conf}_{V \setminus \{v_1, v_2\}, U \setminus \{v_1, v_2\}}(\mathbb{R}^3, K) \times \text{Conf}_2(\mathcal{TR}^3) \quad (4.30)$$

Φ is identity on $\text{Conf}_2(\mathcal{TR}^3) = \mathbb{S}^2$. Φ is the same up to a sign on the faces with respect to three graphs $\Gamma_I, \Gamma_H, \Gamma_X$ in IHX relation, and the orientation guarantees that $I(K, \Gamma_I) = -I(K, \Gamma_H) = I(K, \Gamma_X)$, $\Gamma_I - \Gamma_H + \Gamma_X = 0$ in $\mathcal{A}$.

2. A face where two trivalent vertices are not connected by a dashed line collide is mapped into a submanifold with codimension at least 3 in $(\mathbb{S}^2)^E$ by Φ . Thus, the pullback of any top form on this face must be 0.

3. A face in Γ_S where a trivalent vertex collide with a univalent vertex connected by a dashed line is canceled by another two faces where 2 adjacent univalent collides in some graph Γ_S and Γ_U that appears in a STU relation. Indeed, the first face is again

$$C_{V \setminus \{u,t\}, U \setminus \{u,t\}}(K, \Gamma_S) \times \text{Conf}_2(\mathcal{TR}^3) \quad (4.31)$$

with Φ being identity on $\text{Conf}_2(\mathcal{TR}^3) = \mathbb{S}^2$, while for Γ_T and Γ_U being

$$C_{V \setminus \{u_1, u_2\}, U \setminus \{u_1, u_2\}}(K, \Gamma) \times \{\pm\} \quad (4.32)$$

where $\pm$ are components of $\text{Conf}_2(\mathcal{TR}) = \mathbb{S}^0$. A sign checking shows that these three integrals satisfy $I(K, \Gamma_S) = -I(K, \Gamma_T) = I(K, \Gamma_U)$, using $\Gamma_S - \Gamma_T + \Gamma_U = 0$, these integrals cancel each other.

□

Bott and Taubes also showed by some dimension-counting methods. that

Proposition 4.4.2. *Hidden faces vanish except for anomalous faces.*

Although the anomalous faces do not vanish in general, their contribution to the variation is predictable by

$$\frac{d_a I(K^t, \Gamma)}{dt} = \frac{f_\Gamma}{2} \frac{dI(K^t, \text{---}\bigcirc\text{---})}{dt} \quad (4.33)$$

where f_Γ is a constant that vanishes on even diagrams and non-primitive diagrams. Also, Altschuler and Freidel computed in [1] that

$$\log Z_{CS}(K) = \sum_{\Gamma} \frac{I(K, \Gamma)}{\#\text{Aut}\Gamma} P(\Gamma) \quad (4.34)$$

using the ideas in [6], where P is the projection onto the primitive space $\mathcal{P}(\mathcal{A})$ in lemma 2.4.2. This also implies an analogue of the group-like property Kontsevich integral proposition 3.4.4

Corollary 4.4.3. *$Z_{CS}(K)$ is group-like in $\hat{\mathcal{A}}$.*

Based on proposition 4.4.1, proposition 4.4.2, eq. (4.33) and eq. (4.34), we immediately get

$$\frac{dZ_{CS}(K^t)}{dt} = \left(\sum_{\Gamma} \frac{f_\Gamma}{2\#\text{Aut}\Gamma} \frac{dI(K^t, \text{---}\bigcirc\text{---})}{dt} \Gamma \right) Z_{CS}(K^t) \quad (4.35)$$

The constant

$$\alpha := \frac{1}{2} \sum_{\Gamma \text{ odd primitive}} \frac{f_\Gamma}{\#\text{Aut}\Gamma} \Gamma \quad (4.36)$$

in $\hat{\mathcal{A}}$ is known as the anomaly.

Proposition 4.4.4 (Altschuler-Freidel). *We have the formula of variation of Z_{CS}*

$$\frac{d}{dt}Z_{CS}(K^t) = \left(\frac{d}{dt}I(K^t, \bigcirc \dashrightarrow) \alpha \right) Z_{CS}(K^t) \quad (4.37)$$

where $\alpha \in \hat{\mathcal{A}}$ is a constant called anomaly and it contains only primitive diagrams of odd degree. As a result, there is a formula

$$Z_{CS}(K) = Z(K) \exp \left(I(K, \bigcirc \dashrightarrow) \alpha \right) \quad (4.38)$$

where $Z(K)$ is an isotopy invariant of knot.

These three propositions combined together give the invariance of Z for knot embeddings whose Gauss Integral $I(K^t, \bigcirc \dashrightarrow) = 0$.

Theorem 4.4.5. *Z is a universal finite type invariant*

Proof. If K is an n -singular knot with labeling set J , and $f : J \rightarrow \{\pm\}$, as usual K_f is the knot obtained by desingularization with crossings determined by f . Consider an almost horizontal projection of K with crossings lying in disjoint neighborhoods O_j of ϵ , (the arcs of the crossing is a_j, b_j) and modify it in a neighborhood of diameter ϵ on a_j so that $K_f(\epsilon)$ has zero writhe. Let

$$I(K, \epsilon, \Gamma) = \sum_f (-1)^{\#f^{-1}(-)} I(K_f(\epsilon), \Gamma) \quad (4.39)$$

The configurations that map no univalent vertices of Γ to some a_j do not contribute to $I(K, \epsilon, \Gamma)$ since $I(K_f)$ cancel with $I(K_{f'})$ with $f'(j) = -f(j)$. Thus, we only consider configurations with at least one univalent vertex u_j on a_j . Similarly, the contributions to $I(K, \epsilon, \Gamma)$ of configurations where O_j doesn't contain any other vertex u'_j related by an edge to u_j approaches 0 as ϵ tends to 0.

Thus, we only consider Γ with vertices u_j, u'_j on a_j, b_j , but the only diagram that remains is the chord diagram D_K of K . The number of embeddings up to isotopy with neighborhoods of the double points of the vertices of this chord diagram D_K that respect the pairing of the chords is precisely $\#\text{Aut}(D_K)$, and the integral at each double points tends to 1 as $\epsilon \rightarrow 0$. This completes the proof of the theorem. $\square$

Appendix A

Bialgebra, Filtration and Gradation

In this section, I list several basic results about bialgebras, gradation and filtration. Details can be found in [17] and [9].

An ascending(descending) filtration on a vector space V is an increasing(decreasing) sequence of subspaces $\{V_n\}_{n \in \mathbb{Z}}$. A filtration determines a topological vector space structure on V , which is Hausdorff iff $\bigcap_k V_k = 0$ and discrete if some $V_k = 0$. We say V is exhausted if $\bigcup_k V_k = V$. A map $f : A \rightarrow B$ is continuous iff f preserves the filtration. The Hausdorff completion $V \rightarrow \widehat{V}$ of V is $V \rightarrow \varprojlim V/V_n$, and if this is an isomorphism we say V is complete. The associated graded space of an ascending filtered space V is defined by $\text{gr}V := \bigoplus_k V_k/V_{k-1}$. V is said to have finite type if each V_k is finite dimensional, the filtered dual of a finite type V is the $V^* = \bigcup_k W_k$ with filtration $\{W_k\}_{k \in \mathbb{Z}}$ where W_k consists of functions on V that vanishes on V_k .

It's easy to verify the following facts that $\text{gr}(\bigoplus A_n) = \bigoplus A_n$; $\text{grgr}A = \text{gr}A$. Also $\text{gr}A = \text{gr}(\widehat{A})$, for example $\text{gr} \prod_{k \geq 0} A_k = \bigoplus_{k \geq 0} A_k$, and $(\widehat{A})^* = (A)^*$.

Proposition A.0.1 (lifting criterion). *Suppose $f : A \rightarrow B$ is a filtered map and $\text{gr}f : \text{gr}A \rightarrow \text{gr}B$ is an isomorphism. If $A_0 = A$, then $\bar{f} : A/A_n \rightarrow B/B_n$ is also an isomorphism. If the filtrations are discrete, then $f : A_n \rightarrow B_n$ is an isomorphism. As a consequence, if the filtrations are exhaustive and complete (or in particular if they are exhaustive and discrete), then $\text{gr}f$ lifts to an isomorphism $f : A \rightarrow B$.*

Proof. We prove by induction. Without loss of generality, we assume the filtration is descending. The base is clear. $f : A/A_k \rightarrow B/B_k$ and $f : A_k/A_{k+1} \rightarrow B_k/B_{k+1}$ are isomorphisms implies that $f : A/A_{k+1} \rightarrow B/B_{k+1}$ is isomorphism. The same argument shows that $f : A_n \rightarrow B_n$ is isomorphism in the second case. $\square$

A filtered algebra is an algebra (A, m, u) whose filtration $\{A_k\}_{k \in \mathbb{Z}}$ satisfies $m(A_k \otimes A_l) \leq A_{k+l}$ and $u(A) \subset A_0$. A filtered coalgebra is a coalgebra (A, δ, ϵ) with filtration $\{A_k\}_{k \in \mathbb{Z}}$ such that $\delta(A_n) = \sum_{k+l=n} A_k \otimes A_l$. A filtered bialgebra is a bialgebra that is filtered both as an algebra and a coalgebra with respect to the same filtration. Similarly, a graded bialgebra is a graded vector space $A = \bigoplus_k A_k$ with bialgebra structure satisfies $m(A_k \otimes A_l) \leq A_{k+l}$, $\delta(A_n) \subset \sum_{k+l=n} A_k \otimes A_l$, $u(A) \in A_0$ and ϵ vanishes on A_k for $k \neq 0$. It's easy to verify that if A is a filtered bialgebra of finite type(all factors $A_k/A_{k \pm 1}$ are finite dimensional), then A^* is also a filtered bialgebra with operation induced by dual of operations of A . Also, if A is a filtered bialgebra, then $\text{gr}A$ is also a graded bialgebra.

If A is a filtered bialgebra, the primitive elements of A^* are exactly additive linear functions $f \in A^*$ satisfying $f(ab) = f(a) + f(b)$, $a, b \in A$ and group-like elements are multiplicative linear functions $f \in A^*$ satisfying $f(ab) = f(a)f(b)$, $a, b \in A$.

A graded bialgebra $A = \bigoplus_k A_k$ is said to be connected if u is onto A_0 . A connected graded bialgebra of finite type has a natural Hopf algebra structure. In fact, we have the structure theorem:

Theorem A.0.2 (Milnor-Moore). *Every connected graded cocommutative Hopf bialgebra H is isomorphic to symmetric algebra on primitive subspace $\mathcal{P}(H)$.*

Although the primitive subspace is always non-trivial in such Hopf algebra, the only group-like element is the unit. But, the graded completion of H has a rich set of group-like elements. In fact, there is a one-to-one correspondence between group-like elements and primitive elements via $\log$ and $\exp$.

Appendix B

Intersection of Hyperplanes

Let V be a vector space and $V_1, V_2, \dots, V_n$ be subspaces, let

$$\Delta = \{(v_1, \dots, v_n) \in V^n : v_1 = \dots = v_n\} \quad (\text{B.1})$$

we say $\{V_j\}_{j=1}^n$ intersect generically if the map

$$f : \bigoplus_i V_i \rightarrow V/\Delta \quad (\text{B.2})$$

induced by inclusions is surjective. This is because f is surjective means if we move these planes V_i by ϵ_i , then there is still some $v_i \in V_i$ and $v \in V$ so that

$$v_i + \epsilon_i = v \quad (\text{B.3})$$

for every $i \geq 1$. Equivalently, $\{V_j\}_{j=1}^n$ intersect generically if and only if

$$V_k + \bigcap_{j \neq k} V_j = V \quad (\text{B.4})$$

for every k .

Proposition B.0.1. $V_1, \dots, V_n$ intersect generically if and only if there is a direct sum decomposition

$$V = U' \oplus U_1 \oplus \dots \oplus U_n \quad (\text{B.5})$$

so that

$$V_i = U' \oplus U_1 \oplus \dots \oplus \widehat{U_i} \oplus \dots \oplus U_n \quad (\text{B.6})$$

for every i

Proof. Suppose $\{V_j\}_{j=1}^n$ intersect generically, let $U' = \bigcap_i V_i$. Choose any complement U_i of U' in $\bigcap_{j \neq i} V_j$. Clearly $V = U_i \oplus V_i$. The sum $U' + \sum_j U_j$ is direct because $U' + \sum_{j \neq i} U_j \subset V_i$.

Note that $U_{k+1} \subset \bigcap_{j \leq k} V_j$,

$$\bigcap_{j \leq k} V_j = \bigcap_{j \leq k} V_j \cap (U_{k+1} \oplus V_{k+1}) = U_{k+1} \oplus \bigcap_{j \leq k+1} V_j \quad (\text{B.7})$$

we can inductively write

$$V_k = U' \oplus \bigoplus_{j \neq k} U_j \tag{B.8}$$

and

$$V = U' \oplus U_1 \oplus \cdots \oplus U_n \tag{B.9}$$

□

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